# Salient Is Not Enough: Keeping Every Time Anchor for Temporal Grounding under a Visual-Token Budget

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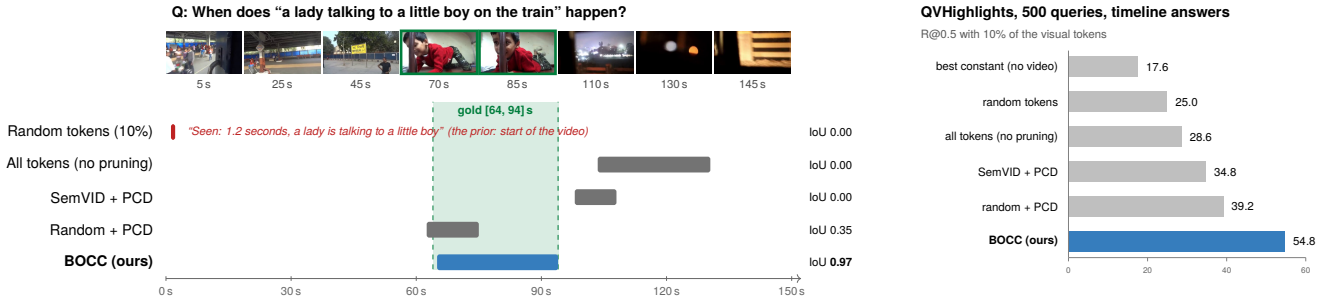



Figure 1. **Temporal grounding from a pruned video memory.** Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers “1.2 seconds”; QA-oriented pruning and the uncompressed model pick a later, wrong scene. BOCC keeps every frame anchored in time and contrasts the answer against the model’s no-video prediction, and recovers the interval (IoU 0.97). Right: R@0.5 over 500 queries when every method answers with a timeline of observations, the format of grounded QA. In this format the prior writes many of the times and BOCC is far above the uncompressed model and the best video-free constant. With a one-line answer the uncompressed model is much stronger, and Coverage alone matches it with a quarter of the tokens (Sec. 6.2).

## Abstract

*Video LLMs are increasingly asked not only* what *happened but* when. *A grounded question such as “how long did I use the paper cutter?” needs both an answer and the time interval of its evidence. To keep long videos tractable, the visual memory of these models is pruned to a small fraction of its tokens, and published pruners lose little multiple-choice accuracy at 10% of the tokens. We show that temporal grounding needs something different:* pruning for QA keeps what is salient, but grounding needs what is everywhere. *A video LLM reads time from timestamps, and a timestamp survives only if some visual token of its frame survives. Pruners that select by saliency, uniqueness or text similarity empty whole frames, delete their timestamps, and fall below random token selection on grounding. When the answer lists observed moments before giving the interval, a video-free prior fills the gaps: asked with no video at all, the model still starts its intervals at the beginning of the window. We propose BOCC, a training-free method that selects frames by the question (BOLT) and adds two parts.* Coverage *keeps, in every layer, the tokens that the actual question attends to most, with an equal per-frame quota so that every time anchor survives. In the settings we tested, a global top-k on the same question attention does as well without the quota, so the question score is what drives the gain. Prior-contrastive decoding (PCD) writes each time value against the model’s own no-video prediction. Because the two parts behave differently across answer formats, we study the format directly. With the standard one-line grounding prompt on the full Charades-STA and QVHighlights sets of TimeLens-Bench, Coverage at 31% of the tokens stays within 0.3 mIoU of the uncompressed model, for both a general and a grounding-tuned model of the same family. It beats random selection by $+0.5$ to $+1.6$ and a lower-resolution input of the same budget by $+2.1$ to $+2.6$, while PCD has no effect on a 513-query subset. In reasoning-style grounded QA (S-EMBER), PCD is the part that matters. With 10% of the tokens, BOCC improves on the uncompressed model by $+11.2$ and $+10.7$ mIoU under the same prompt, although an input restricted to the gold evidence still scores higher in accuracy, and on every metric for the grounding-tuned model. Asking for a timeline of observations instead of a single interval costs single-moment grounding about 30 mIoU, so the answer for-*

*mat is a major variable when evaluating compressed memories. We compare every method with video-free floors and report where the method helps less. On short, densely sampled videos Coverage gains little on grounding (+0.5 mIoU on Charades-STA; NExT-GQA), although on NExT-GQA it keeps the accuracy of the uncompressed model at 10% of the tokens while random selection loses about four points. PCD helps only where the answer format lets the prior choose the times, and it lowers the scored grounding where it disturbs that format.*

## 1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer has two parts, a duration and the interval it comes from. Such *grounded* questions are central to egocentric assistants, video search and step-by-step guidance. What makes them useful also makes them hard. The model must know when things happened across the whole window, including the moments just before and after the event.

At the same time, video LLMs pay for every frame. Qwen3-VL [38] spends about 330 visual tokens per frame (660 per pair of frames), so 64 frames already cost about 21,000 tokens, and long or streaming video multiplies this. Pruning the visual tokens to a small budget has therefore become the default, and a fast-growing literature reports that 10–25% of the tokens are enough to preserve multiple-choice accuracy [14, 18, 27, 35, 41, 44]. In this paper we ask what pruning does to *temporal grounding*. Our answer comes down to one principle with two consequences: **pruning for QA keeps what is salient, but grounding needs what is everywhere.** A QA answer can be read off a few informative frames. An interval can only be read from a timeline, and a timeline needs an anchor in every part of the window.

**Observation 1: QA pruners lose the anchors.** Video LLMs such as Qwen3-VL report time by reading timestamp tokens interleaved with the visual tokens. When all visual tokens of a frame are removed, its timestamp goes too, or it stays with nothing to show. Pruners designed for QA drop frames that look redundant, unremarkable or unrelated to the words of the question, and those are exactly the frames that make a timeline usable. On S-EMBER grounded QA, all four recent pruners we evaluate fall below random token selection at the same budget, and the same holds on QVHighlights and Charades-STA (Sec. 6.3). Random selection is a strong grounding baseline because it leaves a few tokens on every frame. In fact, one of the published scores improves by $5.3^{*}$ mIoU just by ranking tokens within each frame instead of globally (Sec. 3.2).

**Observation 2: where the memory is silent, a prior answers.** Grounded QA invites a reasoning-style answer in which the model lists what it saw and when, and then states the interval. If we ask this way with no video at all, the model still produces confident intervals, and 90% of them start in the first 5% of the window. Pruned memories show the same habit: 30–56% of their predicted intervals start in the first 5% of the window, compared with 16% of the gold intervals, and removing the frames that contain the evidence raises this share from 37% to 51% (Sec. 3.3). So the prior does not belong to any particular pruner. It is what the decoder writes when the memory has nothing to say.

**Observation 3: the answer format decides which failure dominates.** These two observations call for two remedies, and the remedies do not matter equally everywhere. On the same QVHighlights queries and with the same memory, asking for a timeline of observations instead of a single interval costs about 30 mIoU (Sec. 6.2). With the one-line format, selecting tokens by the question helps and contrastive decoding does nothing. With the timeline format, the video-free prior writes many of the times, so contrasting it away helps, as long as the contrast does not break the answer format. For this reason we treat the answer format as an experimental variable instead of a fixed detail of the protocol.

**Observation 4: benchmark floors are high.** Grounding benchmarks have strong positional regularities [37]. On S-EMBER, questions are asked mid-stream and two thirds of the gold intervals end within the last fifth of the window. As a result, the best *constant* interval $[0.15q, q]$ reaches 36.9 mIoU, which is more than the uncompressed model gets when it answers in the timeline format (28.0). Following Otani et al. [37], we report video-free floors for every method (Sec. 3.4).

We propose BOCC (BOLT + Coverage + Contrastive decoding), a training-free method with one component for each failure. An optional BOLT front end chooses which frames to encode (Fig. 3).

- **Coverage** (Sec. 4.1) keeps the anchors. Once the frames are encoded, we prune the KV cache layer by layer. Each frame gets an equal share of the layer's budget, and within each frame we keep the tokens that the actual question attends to most. A timestamp survives exactly when one of its frame's visual tokens survives, so the equal quota guarantees every time anchor while the question decides what each anchor shows. In the settings we tested, a global top-$k$ over the same question attention does as well without the quota. The question score is therefore what drives the gain, and the quota acts as a safeguard for cases where scores collapse onto a few frames.
- **Prior-contrastive decoding (PCD)** (Sec. 4.2) removes the prior where it writes the answer. We choose each time value by contrasting the model's prediction with its prediction from a video-free memory. This is the content-free contrast used in contextual calibration and context-aware

decoding [42, 52], applied only to time values and with a strength that fades where the model is already confident.

Neither operator is new on its own. Per-frame budgets appear in VidCom$^2$, KTV and AdaTP [1, 12, 35], question-attention KV eviction in SnapKV and PruneVid [27, 32], and contrast against a content-free or degraded input in contrastive decoding, classifier-free guidance and VCD [19, 30, 31, 40, 42]. What we contribute is an account of *why* temporal grounding needs each of them and when each one matters, together with a way to make them work together:

1. **A diagnosis of grounding under compression** (Sec. 3). QA-oriented pruners empty frames and fall below random selection. In reasoning-style answers the decoder's video-free prior fills the gaps, and an evidence-removal test shows this causally.
2. **Coverage**, per-layer question-driven token selection under an equal quota (Sec. 4.1). With the standard one-line prompt, on the full Charades-STA and QVHighlights sets and on both Qwen3-VL-8B and the grounding-tuned TimeLens-8B, it matches the uncompressed model with 24% of the visual tokens. It also beats random selection ($+0.5^*$ to $+1.6^*$ mIoU) and a matched low-resolution input ($+2.1^*$ to $+2.6^*$) (Sec. 6.2). On NExT-GQA at 10% of the tokens it keeps the accuracy of the uncompressed model, while random selection loses about four points ($+3.4^*$ and $+3.2^*$ over random). With BOLT frames it exceeds a 1 fps input of similar budget by $+9.2^*$ and $+6.5^*$ mIoU. Random tokens on the same frames, however, already gain $+7.0^*$ and $+6.1^*$, so most of that gain comes from the front end.
3. **PCD and an answer-format study** (Secs. 4.2 and 6.2). On S-EMBER, PCD helps reasoning-style answers on all eight memories we test and on both models. With 10% of the tokens, BOCC improves on the uncompressed model by $+11.2^*$ and $+10.7^*$ mIoU under the same prompt. The same study shows that the answer format alone moves single-moment grounding by about 30 mIoU and decides whether contrastive decoding helps. PCD does nothing for one-line answers, and on NExT-GQA it lowers the scored grounding where it disturbs the format (TimeLens-8B timeline answers, minutes:seconds under the official prompt).
4. **Floor-adjusted, reproducible evaluation** (Sec. 5). We give video-free floors and a length-rescaling baseline for every method, paired bootstrap tests, a reproduction of the published TimeLens-Bench numbers within 0.3 mIoU, and sanity checks with no video and with shuffled frames or timestamps. We also report a broad set of negative results, including the limits of Coverage on short, densely sampled videos and of PCD outside the timeline format.

## 2. Related Work

**Frame selection.** BOLT [34] scores frames against the query with a contrastive image–text model and then picks them by inverse transform sampling. AKS [43], MDP3 [5], FOCUS [2] and Q-Frame [6] optimise relevance jointly with coverage or diversity. All of them are evaluated on multiple-choice QA. Our frame selection front end (Sec. 4.3) adapts BOLT to grounding, and we find that it adds little once the tokens are chosen by the question.

**Visual token pruning for video LLMs.** Query-agnostic methods compress the video by redundancy or saliency. FastVID [41] segments the video and keeps density peaks, VidCom$^2$ [35] sets each frame's budget by how unique the frame is, and FlashVID [18] selects tokens by [CLS] attention and merges them across frames with trees. DyCoke [44] and FrameFusion [21] merge similar tokens. Query-aware methods also use the text. SemVID [14] scores tokens by their word-embedding relevance with MMR diversity, AdaTP [1] boosts segments that resemble the text, LGTTP [4] reallocates tokens according to temporal words in the question, and PruneVid [27] keeps the top 40% of tokens by question attention at a single LLM layer. KTV [12] is the closest in structure, since it selects keyframes first and then tokens under a relevance-weighted per-frame budget. LGTTP is the only method we know of that evaluates grounding, and it does so at a milder budget and offline. BOCC differs from these methods in three ways. Its per-frame quota is equal by design, its score is the attention of the actual question propagated exactly through all layers, and the quota is there to keep every timestamp. We show that emptying frames hurts grounding. We also show that once the actual question's attention is used, the quota acts as a safeguard and is not where the gain comes from (Sec. 6.4).

**KV-cache compression and video memory.** In long-context LLMs, SnapKV [32], $H_2O$ [51], PyramidKV [15] and Ada-KV [20] evict KV entries by attention. Our token stage is SnapKV-style eviction applied to video, with a per-frame constraint added. For streaming video, HERMES [10] compresses the cache with generic pseudo-queries, ReKV [17] retrieves KV blocks when the question arrives, and StreamMem [8] and InfiniPot-V [3] maintain bounded memories. These works report QA only. We build our memory for each question from the frames seen up to the question time. It is therefore causal, but it is not a persistent streaming memory, and we make no claim about the cost of building it (Sec. 7).

**Contrastive decoding and prior correction.** Contrasting a model's prediction with that of a weaker or less informed model is an established decoding tool. Contrastive Decoding [31] subtracts a small "amateur" model and introduced the plausibility constraint that we use. Classifier-free guidance for language models [40] and context-aware de-

coding [42] amplify the difference between the predictions made with and without the conditioning context, which is the form of our Eq. (4). The idea of measuring the prior with a content-free input and correcting for it is older still. Contextual calibration [52] rescales label probabilities using the model's answer to "N/A", and domain-conditional PMI [24] reweighs answers by their task-conditioned prior. For vision-language models, VCD [30] contrasts against a noised image, ICD [46] against a disturbed instruction and M3ID [19] against the absence of the image, while PAI [33] amplifies image attention and subtracts text-only logits. SEASON [7] carries the idea over to video with temporally corrupted negatives. These methods target hallucination, faithfulness or classification. PCD uses the same contrast for another purpose. We apply it only to *time values*, with a strength that adapts to the model's confidence, and we study when it helps and when it does not (Secs. 6.2 and 6.5). This includes checking whether its benefit goes beyond a uniform lengthening of the interval (Sec. 6.5).

**Grounding-tuned video LLMs.** TimeChat [39], VTimeLLM [25] and LITA [26] train video LLMs to localise moments using timestamp-aware encoders, boundary-aware training stages or dedicated time tokens. TimeLens [50] re-annotates the standard grounding benchmarks and fine-tunes Qwen-VL models with reinforcement learning. All of these works improve grounding with a full visual memory. We use TimeLens-8B as our grounding-tuned model and TimeLens-Bench as the standard protocol, and we ask what happens to such a model when its memory is pruned.

**Training-free temporal grounding.** NumPro [48] draws frame numbers onto the frames, FV-Action [9] scans the video with yes/no questions, and REZE [13] recasts grounding as recognition. [Authors to verify] [11] read intervals from calibrated attention heads and debias them against a run on a blank video. All of these methods assume the full set of video tokens. BOCC addresses the complementary problem of grounding from a *compressed* memory, and these readouts could run on top of it **[TODO: run FV-Action and the attention readout on the same 10% memory]**.

**Benchmark biases.** Otani et al. [37] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because the annotations carry positional priors. We find the same on the datasets we use. We therefore adopt their blind baselines as video-free floors in every comparison, and we add a length-rescaling floor for methods that lengthen intervals.

# 3. Grounding under a Token Budget: a Diagnosis

We first establish Observations 1, 2 and 4 of the introduction on S-EMBER, before introducing any method. Observation 3, which concerns the answer format, is studied in Sec. 6.2. Unless stated otherwise, the reference memory in this section is random 10% pruning.

## 3.1. Setting

A question $x$ is asked at time $q$ about a video $v_{0:q}$. The model sees $F{=}64$ frames from $[0, q]$, which are encoded into a KV cache $\mathcal{M}$ of $N$ visual tokens. Timestamp tokens are interleaved with the visual tokens, and Qwen3-VL uses one "`<t seconds>`" for each pair of frames. A budget $\rho$ (by default $\rho{=}0.1$) limits the number of tokens kept in each layer to $\rho N$, and kept timestamps count toward this budget. The model answers with a short *timeline*:

```
Seen: t_1 seconds, …
Seen: t_k seconds, …
Answer: …
Time: [s, e]
```

and we parse the interval $[s, e]$ from the last line. This reasoning-style format suits grounded QA, where the answer describes what was seen before it gives the interval, and we use it throughout the diagnosis. Section 6.2 compares it with the one-line format of grounding benchmarks (*"The event happens in s–e seconds"*). Grounding is scored by IoU with the gold interval, reported as mIoU and R@$\{0.3, 0.5, 0.7\}$. Grounded QA also requires the answer to be judged correct, so GQ@0.5 counts answers that are correct and have IoU $\geq 0.5$. We call an answer an *early start* when its interval starts in the first 5% of the window, and a *leak* when it starts early even though the gold evidence starts later.

## 3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom$^2$, FlashVID and SemVID all score below random token selection (Sec. 6.3). Where the kept tokens lie explains why (Tab. 1). HERMES scores tokens by attention plus recency, so it keeps 27.6% of the tokens in the last quarter of the window and only 2.5–4.7% elsewhere. FastVID retains one representative per run of similar frames, and as a result it keeps a token from only 25% of the sampled frames in a median question. Those runs of similar frames are exactly where a continuing event lives. Random selection keeps about 9% everywhere and never empties a frame. Because a timestamp is useful only when some visual content of its frame survives with it, a selector that empties frames leaves holes in the timeline. These holes hurt even when the score itself is sensible. If we rank HERMES's scores within each frame instead of globally, which gives every frame an equal share ("stratified"), mIoU improves by $5.3^*$.

Table 1. **Where pruning keeps tokens.** Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

| Selector | 0–10% | 10–25% | 25–50% | 50–75% | 75–100% | mIoU | Leak |
|---|---|---|---|---|---|---|---|
| HERMES | 4.7 | 3.9 | 2.6 | 2.5 | 27.6 | 18.3 | 49% |
| Stratified | 8.7 | 8.7 | 8.6 | 8.6 | 8.7 | 23.7 | **29%** |
| Random | 8.9 | 8.8 | 8.6 | 8.7 | 8.6 | **25.8** | 30% |

### 3.3. The prior fills the gaps

**The model answers without video.** When we ask every question with an empty visual memory, the model still writes a timeline. Of its first `Seen:` moments, 89% are at 0 s, and 90% of its intervals start in the first 5% of the window, which gives an 88% leak. The median interval is 0.45× the gold length, and 87% of the ends fall more than half an interval too early. We call this habit of starting early and stopping soon the model's *temporal prior*.

**The prior leaks into answers with video.** The same signature appears, more weakly, in every memory (Fig. 2). About 30% of intervals start early at every random budget and also with no pruning at all, compared with 16% of gold intervals. The share rises when coverage is lost, to 49% for HERMES and 37–39% for HERMES in streaming mode. The leak therefore depends on where the memory is silent and not on how many tokens it keeps.

**Removing the evidence causes the leak.** To test causality, we drop the uniform frames that fall inside the gold interval ("evidence"), or the same number of frames elsewhere ("control"), and keep every other frame with its true timestamp (Tab. 2). Without the evidence, the answers do not spread out. They move to the start of the video instead, and the leak rises by 14 points for both the unpruned and the random 10% memory. In the HERMES 10% memory, the leak also tracks how much of the evidence survives. Questions whose evidence keeps under 5% of its tokens leak 56% of the time, compared with 33% when at least 15% survives.

**An instruction does not remove it.** We added "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt. The leak barely moved (29.5% → 28.0%), and grounding did not change (−0.1 mIoU). In this answer format, then, the prior acts in the output distribution and has to be removed there. We do not yet know whether the same prior appears when the model answers with a single interval. In that format PCD has no effect (Sec. 6.2). Those runs omit the answer reserve, which on its own changes S-EMBER grounding by only +0.1. The null effect suggests that the prior decides fewer of the written times there. **[TODO: no-video answers and leak under the one-line and direct-answer prompts.]**

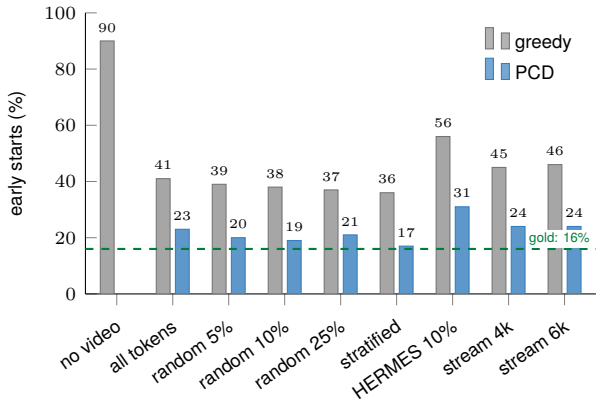



Figure 2. **The temporal prior leaks into every memory, and PCD removes most of it.** Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). PCD roughly halves it everywhere.

Table 2. **Causal test.** Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. PCD: Sec. 4.2.

| | | Greedy | | PCD | |
|---|---|---|---|---|---|
| Memory | Removed | Acc./mIoU/GQ | Leak | Acc./mIoU/GQ | Leak |
| Random 10% | control | 16.6/28.8/8.8 | **31%** | 15.4/34.6/10.5 | **11%** |
| | evidence | 7.8/ 8.6/1.7 | 46% | 9.3/10.3/1.1 | 23% |
| All tokens | control | **19.6/29.4/10.1** | 37% | **18.1/35.9/12.6** | 15% |
| | evidence | 9.1/ 9.4/1.9 | 51% | 9.1/10.5/1.5 | 22% |

**The end is the weak boundary.** Without contrastive decoding, the median end error across memories is 0.40–0.69 of the gold length, while the median start error is 0.14–0.30. The most frequent failure is an interval that lies entirely *before* the gold one, which accounts for 33% of all answers (Tab. 28).

### 3.4. Benchmark floors

Positional regularities make some intervals predictable without the video (Tab. 3). On S-EMBER, questions are asked while the stream is running, and 67% of gold intervals end within the last 20% of the window. As a result, the constant $[0.15q, q]$ scores 36.9 mIoU. Moments in ActivityNet Captions cover 38% of the video on average, and there $[0, 0.95q]$ scores 37.9. On Charades-STA, $[0, 0.45q]$ scores 29.7. QVHighlights is the least biased, with 26.8 mIoU and 17.6 R@0.5. Because these constants are tuned on the test questions, they are an upper bound for any video-free rule of this form. On three of the four datasets, the best constant beats the uncompressed model when it answers with a time-

Table 3. **Video-free floors.** Best constant interval $[aq, bq]$ on a $20\times20$ grid, tuned on the test questions (each metric uses its own best constant; $[a, b]$ is the mIoU optimum), against the uncompressed model. Closed: share of questions with gold end $< 0.8q$.

| | Best constant | | | | | All tokens | |
|---|---|---|---|---|---|---|---|
| Dataset | $[a, b]$ | mIoU | R@0.5 | R@0.7 | Closed | mIoU | R@0.5 |
| S-EMBER | $[.15, 1]$ | 36.9 | 34.7 | 24.2 | 33% | 28.0 | 26.7 |
| QVHighlights | $[0, 1]$ | 26.8 | 17.6 | 8.8 | 62% | 29.1 | 28.6 |
| ActivityNet | $[0, .95]$ | 37.9 | 35.8 | 20.8 | 56% | 22.9 | 18.2 |
| Charades-STA | $[0, .45]$ | 29.7 | 29.6 | 15.4 | 70% | 16.6 | 14.2 |

line. With a one-line answer, Qwen3-VL-8B is well above the constant on ActivityNet and Charades-STA (Sec. 6.2). We therefore report these floors as a reference in every comparison that follows.

# 4. Method: BOCC

BOCC intervenes at the two places where Sec. 3 found the failures. Coverage acts on the memory and prior-contrastive decoding (PCD) acts on the decoder, and both sit behind a frame-selection front end (BOLT). No component needs training, and Fig. 3 gives an overview.

## 4.1. Coverage: equal quotas filled by the question

Once the $F$ frames are encoded, we prune the memory a single time, separately in each of the $L$ layers. Let $\mathcal{T}_f$ be the visual tokens of frame pair $f \in \{1, \dots, F'\}$ and $B$ the per-layer budget for visual tokens.

**Question relevance.** We append the question $x$ to the full, uncompressed memory and run it exactly through all layers, so the queries at layer $\ell$ come from the real hidden states. A visual token $i$ is scored at layer $\ell$ by the attention it receives from the question, averaged over heads $h$ and question positions $j$:

$$a_\ell(i) \;=\; \frac{1}{H\,|x|} \sum_{h=1}^{H} \sum_{j \in x} \mathrm{softmax}_i\Big( \tfrac{1}{\sqrt{d}}\, Q_j^{(\ell,h)}\, K_i^{(\ell,h)^\top} \Big). \tag{1}$$

**Equal quota.** Each frame pair gets the same share of the budget, $B_f = \lfloor B/F' \rfloor$, with any remainder handed out in order. Layer $\ell$ then keeps

$$\mathcal{K}_\ell \;=\; \bigcup_{f=1}^{F'} \mathrm{top}_{B_f} \big\{\, a_\ell(i) \;:\; i \in \mathcal{T}_f \,\big\}. \tag{2}$$

The timestamp tokens of $f$ are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. At any practical budget the equal quota leaves every frame pair with at least one token, so every time anchor survives, while the question decides which part of each frame is shown. Kept tokens keep their original rotary positions, which preserves their temporal order.

**Why equal, and why the question.** We chose the equal quota as a deliberately simple coverage constraint. It guarantees the timeline, which grounding needs and multiple-choice QA does not. It also keeps attention from collapsing onto a few frames. When quotas follow attention mass instead, attention sinks and recency bias can give a single frame up to 227 of the roughly 2,140 kept tokens. In a pre-registered test, relevance-weighted quotas with a floor, computed from SigLIP similarity or from attention mass, did not beat equal quotas ($-0.2$ mIoU). Attention becomes informative once it comes from the *actual* question. Streaming memories such as HERMES [10] score tokens with generic probe questions instead, and generic probes with a global top-$k$ are the weakest variant we measured (18.3 mIoU); adding the equal quota lifts them by $5.3^*$. With the actual question, however, a global top-$k$ without any quota has so far done as well as the equal quota (Sec. 6.4). In that setting the quota acts as a guarantee and is not where the gain comes from.

**Per-layer selection.** Each layer keeps its own set $\mathcal{K}_\ell$. If we force a single shared set by averaging attention over layers, BOCC changes by $-1.5^*$ mIoU, while random selection is unaffected ($+0.7$, n.s.). Per-layer selection therefore helps question-driven scores specifically, and it cannot explain why random selection beats input-level pruners.

**Frames and budget.** An equal quota only works if each frame gets enough tokens to show something. With 2–4 tokens per frame pair, the most-attended tokens carry little content and Coverage collapses (Sec. 6.7). For small budgets we therefore spend the tokens on fewer frames. At a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. Our rule is to use $F{=}64$ when each frame pair keeps at least $\sim$8 tokens and to halve $F$ otherwise. Kept timestamps count against the budget, and at budgets of a few hundred tokens they take up a large part of it, so some of the collapse may come from this accounting (Sec. 6.6). **[TODO: this rule was derived from the results of Sec. 6.7; validate it on held-out data and with the fixed timestamp rule.]**

**Cost.** For each question, Coverage adds one forward pass of the question over the uncompressed memory. Because the full memory is prefilled first, Coverage shrinks the memory that the decoder reads but does not lower the cost of building it. PCD decodes a second, prompt-only row in lockstep at every step, even though its scores are used only inside time values. This roughly doubles the decoder's per-step work, on a much shorter context. **[TODO: measured prefill, time to first token, total latency, peak memory and KV size; the current implementation runs 3–4 scoring passes, reducible to one.]**

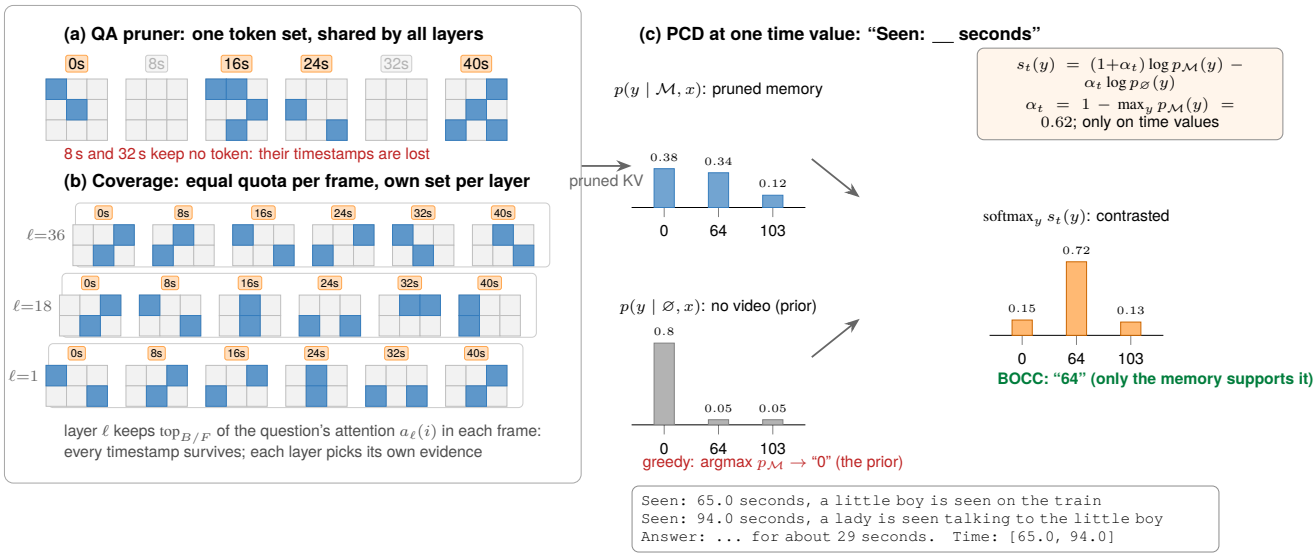



Figure 3. **Overview of BOCC. (a)** QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. **(b)** Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most *in that layer*. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 17). **(c)** PCD at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. Output shown is the real BOCC answer.

### 4.2. PCD: prior-contrastive decoding of time values

Let $p_{\mathcal{M}}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_{\varnothing}(y) = p(y \mid \varnothing, x, y_{<t})$ the distribution with the same prompt and generated prefix but no visual tokens at all. Then

$$\log p_{\mathcal{M}}(y) \;=\; \log p_{\varnothing}(y) \;+\; \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \qquad (3)$$

so when the memory says nothing about $y$, the evidence term is flat and the video-free prior decides. This is the failure described in Sec. 3.3. Prior-contrastive decoding (PCD) amplifies the evidence term,

$$\begin{aligned} s_t(y) \;&=\; (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) \\ &=\; \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \end{aligned} \qquad (4)$$

and picks $y_t = \arg\max_{y \in \mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with $\beta$=0.1, as in VCD [30], so that tokens the memory makes implausible are never promoted. We adapt this rule to grounding in three ways.

**No-video counterfactual.** We use an empty negative memory instead of a noised or shuffled one. This isolates exactly the prior measured in Sec. 3.3, and it is cheap, since its cache holds only the prompt.

**Time scope.** PCD is active only while the model writes a time value, that is, inside a `Seen:` time before its description and inside `Time: [s, e]`. Every other token, including the answer text, is decoded greedily from $p_{\mathcal{M}}$, so PCD leaves the answer alone wherever the memory already provides it.

**Confidence-adaptive strength.** We set $\alpha_t = \alpha_{\max}\left(1 - \max_y p_{\mathcal{M}}(y)\right)$ with $\alpha_{\max}$=1. When the memory makes the model confident, *e.g.* because the evidence really starts at 0 s, PCD fades out and the early answer stands. When the model hesitates between several times, the evidence term decides.

The prompt format matters here. Under the timeline prompt, PCD first acts on the listed `Seen:` moments and then on the interval, so the interval is written from observations that have already been contrasted. With PCD we also reserve 64 tokens of the generation budget for the final `Answer:` and `Time:` lines: once 64 tokens remain and no `Answer:` line has started, the decoder begins one. By itself the reserve changes S-EMBER mIoU by +0.1 (n.s.) and repairs only three truncated answers. When the answer is a single interval, PCD has little to correct. With the one-line grounding prompt it changes mIoU by −0.1 (n.s., Sec. 6.2), in runs without the reserve. For the official direct-answer prompt we do not yet have a clean PCD-only comparison on S-EMBER. On NExT-GQA, PCD under that prompt lowers the scored grounding, because more times are written as minutes:seconds (Sec. 6.5).

### 4.3. BOLT: question-guided frames

When the window is much longer than $F$ frames, the choice of frames to encode can matter as much as the choice of tokens to keep. Our front end samples frames as in BOLT [34]. We embed each second of video once with SigLIP 2 [45], which gives a causal 1 fps index. The relevance $z_j$ of second $j \le q$ is its z-scored cosine similarity with the question, and the $F$ frames are taken at evenly spaced quantiles of

$$p(j) \;=\; \tfrac{1}{2} \cdot \tfrac{1}{q} \;+\; \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \qquad (5)$$

so half of the frames remain uniform. We skip BOLT when uniform frames would be less than 1 s apart. Once the question chooses the tokens, BOLT adds only $+0.8$ to $+0.9$ mIoU, which is not significant on S-EMBER, QVHighlights or ActivityNet. The runs in the main tables include it.

## 5. Experimental Setup

**Models.** Our main model is Qwen3-VL-8B-Instruct [38]. It sees 64 frames at native resolution, sampled uniformly from $[0, q]$ or chosen by BOLT (Sec. 4.3), and decodes greedily or with PCD, at batch size 8 for pruned memories. On S-EMBER we generate at most 384 new tokens with repetition penalty 1.1. On TimeLens-Bench the limit is 512, and on NExT-GQA it is 256 for the two-query setting or 1,024 for the joint answer, both with repetition penalty 1.0. Unless stated otherwise, all methods share the timeline prompt. In Sec. 6.2 we add the one-line prompt of TimeLens-Bench and TimeLens-8B [50], a Qwen3-VL-8B model fine-tuned for grounding. TimeLens-8B frames are capped at about 1.57 MP on S-EMBER and NExT-GQA. On TimeLens-Bench both models use the protocol input (2 fps, at most 14,336 visual tokens and at least 64 per frame), and BOLT runs there use 64 or 128 frames of at most 0.79 MP. Batch size alone changes about half of the greedy outputs and moves a score by up to about two points. Comparisons with the uncompressed model, which runs at batch size 1–2, include this noise.

**Two clusters.** We made the later runs on a second GPU cluster. These include most fixed-rule and all 5% NExT-GQA runs, BOLT on the full QVHighlights set, the official S-EMBER prompt, and the TimeLens-8B gold-informed runs. Some paired differences combine runs from both clusters (marked where they occur). Runs repeated on both clusters agree within noise. NExT-GQA without pruning moves by $+0.4$ mIoU (n.s.), with 469 of 510 replies identical. On S-EMBER, random tokens move by $+0.2$ and BOCC by $-0.9$ mIoU (both n.s.), and the TimeLens script on QVHighlights moves by $-0.2$ (n.s.).

**Datasets.** **S-EMBER** grounded QA [47]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream, so the memory contains only the frames before the question. **QVHighlights** [29] val: 500 queries with a single relevant window, sampled with seed 0 from the 1,020 single-window queries, on videos of about 150 s. **ActivityNet Captions** [28] val_1: 500 queries, one sentence per video, on videos of about 115 s. **Charades-STA** [22] test: 500 queries on videos of about 30 s. For these three datasets the question is *"When does the following happen in the video: <query>?"* and the window is the whole video. **TimeLens-Bench** [50]: the re-annotated Charades-STA, QVHighlights and ActivityNet sets, evaluated with their one-line prompt on the full sets and on a 513-query QVHighlights subset. **NExT-GQA** [49] test: 1,005 questions on videos of about 40 s. The answer and the interval are asked either in two separate queries (multiple choice or open-ended) or, for open-ended answers, in one joint reply. Our QVHighlights and ActivityNet copies are low-resolution re-encodes with 153 and about 84 visual tokens per frame pair, compared with 660 on S-EMBER, so we also report budgets in absolute tokens.

**Budget.** The default budget is 10% of the visual tokens of the 64-frame memory, timestamps included (about 2,140 tokens on S-EMBER, about 520 on QVHighlights). We report ActivityNet at 30% (400–900 tokens), because at 10% its memory holds only 120–300 tokens (Sec. 6.6). Since timestamps count against the budget, a KV selector at a nominal $\rho$ keeps fewer visual tokens than an input pruner at the same $\rho$; on NExT-GQA at 10% it keeps about 3.4% of the visual tokens. A frame's timestamp is kept when the frame keeps a visual token. An earlier version of this rule could let timestamps fill the budget of very small memories and leave no visual token. The NExT-GQA 10% runs use the fixed rule (Sec. 6.2) and the other runs use the earlier one. The difference can matter only where timestamps take a large share of the budget (Sec. 6.6).

**Baselines.** We compare against all tokens (no pruning), random tokens, and four QA pruners: FastVID [41], VidCom$^2$ [35], FlashVID [18] (vision-side stages) and SemVID [14]. The pruners are re-implemented and checked to keep the same tokens as the authors' code on test inputs. Each baseline is run with and without PCD. A global question top-$k$ without quotas is compared in Sec. 6.4. **[TODO: SnapKV-style observation-window eviction, PruneVid, and FV-Action / attention readout on the same memory.]**

**Floor-adjusted evaluation.** For every dataset we report three video-free or video-light references. (i) The *best constant interval* $[aq, bq]$ over a $20\times20$ grid, tuned on the test questions, gives an upper bound for any constant rule. (ii) *Length rescaling* stretches the no-PCD prediction about its start by a factor of 1.76, fitted once on S-EMBER and applied unchanged to the other datasets. It tests whether a gain could come from longer intervals alone. (iii) We split the questions into *closed* ones, whose gold interval ends before $0.8q$, and the rest. End-snapping on S-EMBER is discussed

in Appendix D.

**Metrics and statistics.** We report mIoU and R@$\{0.3, 0.5, 0.7\}$. On S-EMBER we also report answer accuracy and GQ@0.5, judged by a local Qwen3.8-27B-FP8 model with the official S-EMBER judge prompt. Judging was repeated in three sessions, whose verdicts for the same run differ by up to about one accuracy point. We therefore never pair Acc. or GQ@0.5 across sessions, and each table uses a single session, named in its caption: session A for Tab. 6, B for Tabs. 11 and 25, and C for Tab. 26. The early-phase tables listed in Sec. 6 use their own earlier sessions and are compared only within themselves. The same Qwen3-VL-8B BOCC run scores 21.7 accuracy and 13.3 GQ@0.5 in session A and 21.9 and 13.5 in session B. The TimeLens-8B BOCC run scores 27.4 and 17.5 in B and 26.5 and 17.1 in C. Open-ended NExT-GQA answers are judged by the same model with the Video-ChatGPT prompt [36], in separate batches that share a cache. On S-EMBER, answers that give times as minutes:seconds are converted to seconds, for every method alike (Appendix A). The joint NExT-GQA answers are scored with the official S-EMBER parser, which counts such times as missing. We use a paired bootstrap over questions with 2,000 resamples, and we mark with $^*$ the differences whose 95% interval excludes 0. **[TODO: video-clustered bootstrap.]**

**Protocol.** We made all design choices (scores, quotas, $\alpha$, scope) on S-EMBER. We then froze the configuration and ran it once on QVHighlights, ActivityNet and Charades-STA. **[TODO: held-out S-EMBER videos as a test split.]**

# 6. Results

We organise the results by question. Section 6.1 compares BOCC with all baselines and floors on four datasets, using the timeline answer format. Section 6.2 then repeats the key comparisons under the standard one-line prompt, on a grounding-tuned model and on NExT-GQA. Section 6.3 looks at how published pruners behave on grounding, and Secs. 6.4 and 6.5 analyse the two components. Sections 6.6 and 6.7 study the budget and how to spend it. The remaining sections compare with upper bounds that use the gold interval (Sec. 6.8), analyse BOLT (Sec. 6.9), list what did not work (Sec. 6.10) and examine the errors that remain (Sec. 6.11).

**Reading the tables.** Numbers are mIoU, or mIoU / R@0.5 where two are given. $\Delta$ is a paired difference, and $^*$ marks differences whose 95% paired bootstrap interval excludes 0. Tables built from the early S-EMBER phases (Tabs. 18 to 21 and 28 and Tab. 2) use an earlier implementation of PCD, the official parser and batch size 1. On the S-EMBER rows we checked, the two parsers differ by 0.2–0.7 mIoU. The exception is the oracle selector with PCD, where they differ by 1.5 mIoU because 17 answers are written in minutes:seconds. Batch size changes no paired result. All other S-EMBER tables use the minutes:seconds-aware parser. In every table the best value is in **bold** and the second best underlined, per column unless a caption says per row. Floors and differences ($\Delta$) are not ranked.

## 6.1. Main results

Tables 4 and 5 lead to three conclusions.

(i) With the timeline prompt, BOCC nearly doubles the uncompressed model on the least biased benchmark. On QVHighlights with 10% of the tokens, BOCC reaches 54.8 R@0.5 (R@0.7: 30.2). All tokens give 28.6 (18.4), and random tokens 25.0 (16.2). BOCC also clears every floor by a wide margin, with +22.1 mIoU over the best constant and +8.6 over length rescaling. This comparison holds the prompt fixed. The one-line prompt of TimeLens-Bench gives the uncompressed model far higher scores, and there the gain of BOCC is smaller (Sec. 6.2).

(ii) On S-EMBER, the gain holds where the window end is not a shortcut. BOCC beats all tokens, random selection and SemVID under the same decoding. Against the floors its pooled gain is small and not significant, because two thirds of S-EMBER intervals end at the question time. Closed intervals do not offer that shortcut, and on them BOCC gains +7.6$^*$ mIoU over the best constant and +7.5$^*$ over length rescaling (Tab. 29).

(iii) On ActivityNet and Charades-STA, the timeline prompt is the bottleneck. With this prompt and our low-resolution copies, the uncompressed model falls far below the constant floor on both datasets (22.9 *vs*. 37.9 and 16.6 *vs*. 29.7). With the one-line prompt of TimeLens-Bench, the same model scores 47.1 and 48.5 mIoU (Sec. 6.2). On ActivityNet, BOCC improves on all tokens and on every pruner, yet no method reaches the floor. On Charades-STA, Coverage is worse than random selection (Sec. 6.7). We count both datasets among the limits of the method.

**Answers.** BOCC also improves the answers (Tab. 6). Its accuracy is 21.7, against 17.3 for random tokens (+4.4$^*$) and 19.8 for all tokens. GQ@0.5 rises to 13.3, from 7.4 and 8.8 respectively. These numbers come from judging session A; in session B the same BOCC run scores 21.9 and 13.5 (Tabs. 11 and 25). Coverage alone gives the best accuracy (22.5), so PCD trades about one point of accuracy for its grounding gain. Differences below about two points of accuracy are within the decoding noise of the judge setup. **[TODO: SemVID + PCD in the same judging session.]**

**Question types.** BOCC improves duration and counting questions most (Tab. 7). Location questions ("where did I put …") remain essentially unsolved, with mIoU $\leq$ 11 for every method. Part of the reason is the labels. For 62 of the 100 location questions, the annotators disagree on the interval (mean pairwise IoU $<$ 0.6), which caps what any method can score.

Table 4. **Grounding with 10% of the visual tokens** (ActivityNet: 30%), mIoU / R@0.5. Grey rows are floors: the best video-free constant interval, tuned on the test questions, and the BOLT + Coverage prediction without PCD lengthened by a factor fitted on S-EMBER. Coverage rows include BOLT, which is skipped on Charades-STA, where uniform frames are already dense. Empty cells: not yet computed. S-EMBER closed intervals: Tab. 29.

| Method | S-EMBER | QVHighlights | ActivityNet (30%) | Charades-STA |
|---|---|---|---|---|
| All tokens (no pruning) | 28.0 / 26.7 | 29.1 / 28.6 | 22.9 / 18.2 | 16.6 / 14.2 |
| Random tokens | 25.6 / 24.6 | 26.9 / 25.0 | 22.9 / 18.6 | 17.4 / 13.8 |
| FastVID [41] | 17.9 / 15.8 | 19.2 / 15.4 | | 13.3 / 11.6 |
| VidCom$^2$ [35] | 20.4 / 18.7 | 19.9 / 18.2 | | 12.3 / 8.4 |
| FlashVID [18] | 22.7 / 21.5 | 22.8 / 20.4 | 20.5 / 15.4$^\dagger$ | 14.1 / 10.8 |
| SemVID [14] | 23.3 / 20.6 | 24.5 / 21.2 | 22.8 / 18.6 | 14.6 / 11.2 |
| BOLT + Coverage | 32.5 / 34.1 | 34.5 / 35.6 | 25.4 / 21.2 | 16.4 / 13.6 |
| BOLT + Coverage + PCD | **39.3 / 40.8** | **48.8 / 54.8** | **34.7 / 32.4** | 23.3 / 21.2 |

$^\dagger$ at 10%.

Table 5. **Paired gains of BOCC in mIoU** ($^*$: 95% paired bootstrap interval excludes 0).

| BOCC *vs.* | S-EMBER | QVHighlights | ActivityNet (30%) | Charades-STA |
|---|---|---|---|---|
| All tokens | $+11.2^*$ | $+19.7^*$ | $+11.8^*$ | |
| Random + PCD | $+4.2^*$ | $+11.0^*$ | $+3.3^*$ | $-2.9^*$ |
| SemVID + PCD | $+4.5^*$ | $+14.1^*$ | $+4.5^*$ | |
| Best constant | $+2.4$ | $+22.1^*$ | $-3.1^*$ | $-6.4^*$ |
| Length rescaling | $+1.0$ | $+8.6^*$ | $+0.8$ | $-0.7$ |

Table 6. **Grounded QA on S-EMBER** (10%; all rows from judging session A, Sec. 5). ΔAcc. *vs.* random tokens.

| Method | Acc. | GQ@0.5 | mIoU | ΔAcc. |
|---|---|---|---|---|
| All tokens | 19.8 | 8.8 | 28.0 | +2.5 |
| Random tokens | 17.3 | 7.4 | 25.6 | – |
| FastVID | 15.2 | 4.8 | 17.9 | −2.1 |
| VidCom$^2$ | 16.2 | 5.9 | 20.4 | −1.1 |
| FlashVID | 17.3 | 6.1 | 22.7 | +0.0 |
| SemVID | 19.8 | 7.6 | 23.3 | +2.5 |
| Coverage, uniform frames | **22.5** | 11.2 | 31.7 | $+5.3^*$ |
| Random + PCD | 20.4 | 12.2 | 35.0 | +3.2 |
| SemVID + PCD$^\circ$ | 17.9 | 9.9 | 32.3 | +0.6 |
| BOCC | 21.7 | 13.3 | **39.3** | $+4.4^*$ |
| BOCC + coverage gate | 22.1 | **13.5** | **39.3** | $+4.8^*$ |

$^\circ$ earlier implementation of PCD.

Table 7. **S-EMBER by question type**, mIoU / R@0.5 (10%).

| Method | Duration (191) | Counting (184) | Location (100) |
|---|---|---|---|
| All tokens | 37.6 / 40.3 | 30.1 / 25.5 | 5.9 / 3.0 |
| Random tokens | 33.2 / 35.1 | 29.6 / 26.1 | 3.5 / 2.0 |
| BOLT + Coverage | 43.2 / 46.6 | 35.3 / 36.4 | 7.1 / 6.0 |
| Random + PCD | 41.0 / 40.8 | 41.9 / 40.2 | **11.0 / 9.0** |
| SemVID + PCD | 38.9 / 36.6 | 44.6 / **50.0** | 8.9 / 8.0 |
| BOCC | **47.6 / 50.8** | **46.9** / 48.4 | 9.3 / 8.0 |

### 6.2. Standard prompt, a grounding-tuned model, and short videos

The results so far use one model and the reasoning-style timeline prompt on every dataset. Here we test the method in three further settings: under the standard one-line grounding prompt of TimeLens-Bench [50], on a grounding-tuned model (TimeLens-8B [50]), and on NExT-GQA [49]. Unless stated otherwise, paired differences are computed on the same questions. Unpruned runs use batch size 1–2 and pruned runs batch size 8, and this difference alone can move scores by up to about two points (Sec. 5). Some runs were made on a second GPU cluster. We flag every paired difference that combines runs from the two clusters; repeated runs across clusters differ by less than one mIoU point, and none of these differences is significant (Sec. 5).

**Our pipeline reproduces published numbers.** With the TimeLens evaluation script, Qwen3-VL-8B scores 48.5, 59.6 and 47.1 mIoU on the full Charades-STA, QVHighlights and ActivityNet sets of TimeLens-Bench, against published scores of 48.3, 59.4 and 46.8. TimeLens-8B scores 55.4, 65.7 and 53.2, against 55.2, 65.5 and 53.2. Our own pipeline without pruning also agrees with the script on the same queries (Tab. 30). The differences are +0.5 mIoU on Charades (400 queries), +1.4 on ActivityNet (400) and 0.0 on QVHighlights (513), and none is significant.

**The timeline prompt is costly for single-moment ground-**

Table 8. **Answer format** (QVHighlights, 513 TimeLens-Bench queries, Qwen3-VL-8B, 300 frames, 31% of the tokens), mIoU. One-line: the TimeLens-Bench prompt. Timeline: the prompt of all other tables. The one-line PCD run used no answer reserve, the timeline PCD runs a reserve of 64 tokens (Sec. 4.2); on S-EMBER the reserve alone changes mIoU by +0.1 (n.s.).

| | One-line prompt | | Timeline prompt | |
|---|---|---|---|---|
| Selection | greedy | PCD | greedy | PCD |
| All tokens | 60.5 | – | 30.4 | – |
| Random tokens | 58.2 | – | **33.4** | 43.9 |
| Coverage | **61.7** | 61.6 | 30.7 | 43.9 |

**ing.** Table 8 changes only the prompt. On QVHighlights queries, asking the model for a timeline of observations before the interval costs about 30 mIoU compared with the one-line prompt: $-30.0^*$ without pruning and $-31.0^*$ for Coverage. PCD recovers part of this loss, adding $+13.2^*$ on Coverage and $+10.4^*$ on random tokens, yet it stays far below the one-line prompt ($-17.8^*$). Under the one-line prompt PCD has no effect ($-0.1$, n.s.), and the same is true with BOLT ($-0.3$ and $-0.5$). The one-line PCD runs used no answer reserve, unlike the timeline runs, so this comparison is not matched on the reserve. We do not expect the reserve to explain the difference, because on S-EMBER the reserve alone does not change grounding ($+0.1$, n.s.). **[TODO: one-line PCD with the 64-token reserve.]** With the timeline prompt, Coverage is not better than random selection on these queries ($-2.8^*$ greedy, $+0.1$ with PCD).

These results bear on the earlier sections in two ways. The QVHighlights, ActivityNet and Charades-STA numbers of Tab. 4 are measured in a format that handicaps the uncompressed model, so the gain over "all tokens" reported there (Sec. 6.1) does not hold against the one-line prompt. And PCD corrects time values in reasoning-style answers, where the model writes several times before the interval. When the answer is a single interval, there is little for it to correct.

**Coverage under the standard prompt.** With the one-line prompt (Tab. 9, top), Coverage at 24% of the visual tokens beats random selection on both models, by $+3.5^*$ and $+2.0^*$ mIoU. It also beats SemVID ($+2.9^*$) and the TimeLens input re-sampled to a similar budget ($+3.7^*$, $+3.0^*$). Against the uncompressed model it is level: $+1.2$ for Qwen3-VL-8B, with a lower bound just at zero, and $+0.2$ for TimeLens-8B.

**BOLT frames on the full QVHighlights set.** On the full QVHighlights set (Tab. 9, bottom), at about 1.2k tokens per query, BOLT with Coverage exceeds a 1 fps TimeLens input capped at 1,365 video tokens by $+9.2^*$ mIoU on Qwen3-VL-8B and $+6.5^*$ on TimeLens-8B. BOLT with random tokens, however, already gains $+7.0^*$ and $+6.1^*$. Coverage adds $+2.2^*$ over random tokens for Qwen3-VL-8B and $+0.4$ (n.s.) for TimeLens-8B, and at 128 frames (about 4.5k tokens) it adds $+0.6$ and $-0.0$ (n.s.). Most of this gain therefore comes from choosing the frames by the question and keeping their resolution, not from Coverage. The comparison also leaves out the cost of the SigLIP index. On the 513-query subset at 897 visual tokens, BOLT + Coverage led a 1,125-token 1 fps input by $+8.8^*$ and $+6.8^*$, and Coverage led random tokens by $+2.8^*$ with Qwen3-VL-8B.

Table 9. **Standard grounding prompt** (QVHighlights, one-line prompt), mIoU / R@0.5. Top: 513 TimeLens-Bench queries, 300 frames at protocol size; Kept: visual tokens in memory (keep ratios count timestamp tokens too: 31% of all tokens is 24% of the visual tokens). Bottom: the full set (1,541 queries) with 64 or 128 frames chosen by BOLT; Kept: tokens per query *including* timestamps. TimeLens input: the TimeLens-Bench input at a lower resolution, capped at the given number of video tokens. Each block is ranked separately.

| Method | Kept | Qwen3-VL-8B | TimeLens-8B |
|---|---|---|---|
| All tokens | 13,650 | 60.5 / 64.9 | 67.1 / 73.1 |
| TimeLens input, 2 fps low-res | 4,200 | 58.0 / 62.0 | 64.3 / 68.8 |
| Random tokens (31%) | 3,271 | 58.2 / 60.8 | 65.3 / 70.4 |
| SemVID (31%) | 4,232 | 58.8 / 62.4 | – |
| Coverage (31%) | 3,271 | **61.7 / 66.7** | **67.3 / 73.9** |
| All tokens, full set | all | 59.6 / 64.5 | 65.7 / 71.8 |
| TimeLens input, 1 fps low-res | ≤1,365 | 48.0 / 51.5 | 57.0 / 61.4 |
| BOLT 64 + random (23%) | 1,192 | 55.0 / 59.8 | 63.1 / 69.7 |
| BOLT 64 + Coverage (23%) | 1,192 | **57.2 / 62.9** | **63.5 / 70.0** |
| TimeLens input, 2 fps low-res | ≤4,500 | 57.3 / 61.6 | 63.3 / 68.4 |
| BOLT 128 + random (43%) | 4,455 | 59.9 / 65.3 | **65.3 / 71.1** |
| BOLT 128 + Coverage (43%) | 4,455 | **60.5 / 66.1** | **65.3** / 70.8 |

Both models lose accuracy against their uncompressed memory at about 1.2k tokens (BOLT + Coverage: $-2.5^*$ and $-2.1^*$ mIoU). At about 4.5k tokens they are level with it ($+0.9$ and $-0.4$, n.s.), and above the 2 fps low-resolution input capped at 4,500 tokens ($+3.2^*$ and $+2.0^*$). Small budgets need fewer frames (Sec. 6.7). With the protocol input instead of BOLT frames, about 1.2k timestamp tokens use up a budget of 8.2%, so no visual token survives and both selectors collapse (4.6 and 4.2 mIoU).

**Full TimeLens-Bench sets.** On the complete Charades-STA (3,363 queries) and QVHighlights (1,541) sets of TimeLens-Bench (Tab. 10), Coverage at 31% of the tokens is lossless on both models, within $-0.2$ to $+0.1$ mIoU (n.s.). It beats random selection in all four cells by $+0.5^*$ to $+1.6^*$, although for Qwen3-VL-8B on Charades-STA the interval only just excludes 0. It also beats, in all four cells, the TimeLens input reduced to the same budget by lowering the resolution ($+2.1^*$ to $+2.6^*$). Random selection, in turn, falls below the uncompressed model in three of the four cells ($-0.9^*$ to $-1.6^*$); the exception is Qwen3-VL-8B on

Table 10. **Full TimeLens-Bench sets** (one-line prompt, 31% of the tokens), mIoU / R@0.5. All tokens and low-res: the TimeLens evaluation script, the latter at a lower resolution with a cap of 3,500 / 4,500 *video* tokens on Charades-STA / QVHighlights. Coverage and random keep 3,348 / 4,652 tokens per query *including* timestamps, so the low-res input sees at least as many visual tokens. Paired ΔmIoU below; † interval lower bound +0.01.

| | Qwen3-VL-8B | | TimeLens-8B | |
|---|---|---|---|---|
| Method | Charades | QVHighlights | Charades | QVHighlights |
| All tokens (no pruning) | 48.5 / **54.0** | 59.6 / **64.5** | **55.4 / 63.1** | **65.7 / 71.8** |
| TimeLens input, low-res | 46.0 / 49.6 | 57.3 / 61.6 | 52.8 / 59.6 | 63.3 / 68.4 |
| Random tokens (31%) | 48.2 / 52.3 | 58.1 / 62.6 | 54.5 / 61.7 | 64.2 / 69.6 |
| Coverage (31%) | **48.6 / 54.0** | **59.7 / 64.5** | 55.2 / **63.1** | 65.4 / 71.0 |
| Coverage *vs.* random | +0.5*† | +1.6* | +0.7* | +1.2* |
| Coverage *vs.* low-res | +2.6* | +2.4* | +2.4* | +2.1* |
| Coverage *vs.* all tokens | +0.1 | +0.1 | −0.2 | −0.2 |

Table 11. **A grounding-tuned model on S-EMBER** (10%, timeline prompt), mIoU / R@0.5 and Acc. / GQ@0.5. All Acc. and GQ@0.5 values are from judging session B (Sec. 5).

| | Qwen3-VL-8B | | TimeLens-8B | |
|---|---|---|---|---|
| Method | mIoU / R@0.5 | Acc. / GQ | mIoU / R@0.5 | Acc. / GQ |
| All tokens (no pruning) | 28.0 / 26.7 | 19.6 / 8.8 | 33.1 / 34.1 | 20.6 / 12.2 |
| Random tokens | 25.6 / 24.6 | 17.1 / 7.2 | 28.2 / 26.7 | 19.6 / 9.5 |
| Random + PCD | 35.0 / 33.9 | 20.0 / 12.0 | 38.4 / 39.4 | 20.4 / 12.4 |
| BOLT + Coverage | 32.5 / 34.1 | **22.3** / 11.2 | 37.2 / 37.9 | 22.9 / 13.1 |
| BOCC | **39.3 / 40.8** | 21.9 / **13.5** | **43.9 / 46.9** | **27.4 / 17.5** |
| BOCC *vs.* all tokens | +11.2* | +2.3 / +4.6* | +10.7* | +6.7* / +5.3* |

Table 12. **NExT-GQA, multiple choice** (test, 1,005 questions, 64 frames, two-query protocol: one query for the answer, one for the interval; no PCD). Coverage and random budgets count timestamp tokens, so at 10% they keep about 3.4% of the visual tokens; SemVID, an input pruner, keeps the nominal share (matched comparison: Tab. 13). All tokens: batch size 2, first cluster; 10% and 5% rows: batch size 8, second cluster. † Earlier timestamp rule, which could leave no visual tokens on low-budget questions (about 282 of 2,010 prompts at 5%); reruns with the fixed rule are pending. All other KV-selector rows use the fixed rule or are unaffected (30%).

| Model | Method | Acc@QA | Acc@GQA | mIoP | mIoU |
|---|---|---|---|---|---|
| Qwen3-VL-8B | All tokens | 79.7 | 38.9 | **48.0** | **37.7** |
| | Random 30% | 78.6 | 38.6 | 47.5 | 37.3 |
| | SemVID 30% | 78.9 | 38.5 | 46.2 | 35.8 |
| | Coverage 30% | **79.8** | **39.4** | 47.9 | 37.6 |
| | Random 10% | 75.7 | 36.7 | 46.6 | 36.4 |
| | Coverage 10% | 79.1 | 38.5 | 47.4 | 37.0 |
| | Random 5%† | 68.3 | 27.3 | 40.5 | 30.5 |
| | SemVID 5% | 74.0 | 30.0 | 41.0 | 30.0 |
| | Coverage 5%† | 75.0 | 32.4 | 41.6 | 31.4 |
| TimeLens-8B | All tokens | **76.7** | **41.5** | **52.1** | 39.8 |
| | Random 10% | 72.8 | 38.6 | 50.1 | 38.9 |
| | Coverage 10% | 76.0 | 40.8 | **52.1** | **40.0** |
| | Random 5%† | 66.0 | 28.7 | 43.6 | 34.0 |
| | Coverage 5%† | 71.5 | 33.3 | 46.0 | 35.6 |

Charades-STA (−0.3, n.s.). The gap to random is largest on QVHighlights, where videos are longer. On Charades-STA, 31% leaves about 3,350 of 10,800 tokens, and random selection is already close. Keeping full resolution and choosing tokens works better than spending the same budget on a lower resolution. Even random selection beats the low-res input on Charades-STA (+2.2* and +1.7*), though not significantly on QVHighlights. **[TODO: full ActivityNet set (paused).]**

**A grounding-tuned model on grounded QA.** S-EMBER answers explain before they ground, so the timeline format is natural there. TimeLens-8B, which is trained for grounding, scores above Qwen3-VL-8B in every row of Tab. 11, and both models rank the methods in the same order. With 10% of the tokens, BOCC improves on the uncompressed TimeLens-8B on every metric: +10.7* mIoU, +6.7* accuracy and +5.3* GQ@0.5. Its lead over random selection is larger, at +15.7* mIoU, +7.8* accuracy and +8.0* GQ@0.5. Each component helps on its own. PCD adds +10.2* mIoU to random tokens and +6.7* to BOLT + Coverage, and BOLT + Coverage adds +9.0* to random tokens.

We repeated the judging in a second session (C), which also scores the gold-informed TimeLens-8B runs of Sec. 6.8. It gives BOCC 26.5 accuracy and 17.1 GQ@0.5, compared with 27.4 and 17.5 in session B. The two sessions agree within 1.1 accuracy points on every TimeLens-8B row, and every difference above keeps its sign and significance; over all tokens, session C gives +6.7* accuracy and +5.5* GQ@0.5. Session C also shows that the grounding-tuned model profits more from the method. In it, with Qwen3-VL-8B rerun on the second cluster, BOCC on TimeLens-8B is above BOCC on Qwen3-VL-8B by +5.4* mIoU, +7.0* accuracy and +5.5* GQ@0.5. With random tokens the two models differ by only +2.4 mIoU (n.s.). TimeLens-8B's own gold-informed upper bounds, however, remain well above BOCC (Sec. 6.8).

**Short videos.** NExT-GQA videos last about 40 s, so 64 frames already sample them densely (Tab. 12). At 10%, Coverage is indistinguishable from the uncompressed model on every metric for both models. For Qwen3-VL-8B the differences are −0.6 Acc@QA, −0.4 Acc@GQA and −0.7 mIoU; for TimeLens-8B they are −0.7, −0.7 and +0.2; none is significant. Random selection, by contrast, loses about four points of accuracy (−4.0* and −3.9*; these pairs combine the two clusters). Coverage therefore beats random selection on Acc@QA, by +3.4* with Qwen3-VL-8B and +3.2* with TimeLens-8B. On grounding the two selectors differ little. Coverage leads by +0.6 mIoU (n.s.) for Qwen3-VL-8B. For TimeLens-8B it leads by +1.9* mIoP and +1.1 mIoU (n.s.), and its +2.2 Acc@GQA has an interval reach-

Table 13. **NExT-GQA checks** (Qwen3-VL-8B, multiple choice, the 269 questions of the first 50 test videos; 10% rows use the fixed timestamp rule). Grey rows remove or scramble the video. SemVID 3.4% and 24.9% keep as many visual tokens as Coverage at 10% and 30%. The 10% block is ranked; the 30% rows are references for the paired comparison in the text.

| Memory | Acc@QA | Acc@GQA | mIoP | mIoU |
|---|---|---|---|---|
| All tokens | 83.6 | 40.1 | 49.0 | 37.6 |
| *No video* | *51.3* | *8.2* | *17.7* | *7.2* |
| *Frames shuffled* | *83.6* | *32.3* | *38.7* | *30.9* |
| *Timestamps shuffled* | *84.0* | *27.5* | *33.3* | *26.6* |
| Random 10% | 80.3 | 37.2 | 46.4 | 34.8 |
| SemVID 10% (nominal) | 78.8 | 35.3 | 44.7 | 32.4 |
| SemVID 3.4% (matched) | 71.4 | 31.2 | 41.4 | 28.9 |
| Coverage 10% | **84.4** | **40.5** | **49.0** | **37.5** |
| SemVID 30% (nominal) | 81.8 | 39.0 | 47.7 | 36.4 |
| SemVID 24.9% (matched) | 81.0 | 37.5 | 45.6 | 34.3 |

Table 14. **NExT-GQA, open-ended** (Qwen3-VL-8B, 1,005 questions, options removed). Answers are judged correct when they match the ground truth in meaning (LLM judge, Video-ChatGPT prompt [36]); Acc@GQA also requires IoP $\geq 0.5$. Two-query: the answer prompt plus the TimeLens grounding query. Joint: one answer with the interval, with the timeline prompt or the official S-EMBER prompt; joint intervals are scored with the official S-EMBER parser, which counts times written as minutes:seconds as missing. Rows come from separate judge batches of the same judge with a shared cache (a rerun of all tokens reproduces Acc@QA within 0.3). PCD uses the 64-token answer reserve. † Earlier timestamp rule (Tab. 12); all other 10% rows use the fixed rule. All-tokens rows: batch size 2, first cluster; two-query and timeline 10% rows: batch size 8, second cluster; official-prompt rows and 30% rows: first cluster. § Rerun on the second cluster, whose own no-PCD rerun scores 33.2 / 17.1 / 48.0 / 37.5. Blocks are ranked separately.

| Protocol | Method | Acc@QA | Acc@GQA | mIoP | mIoU |
|---|---|---|---|---|---|
| Two-query | All tokens | **33.5** | 17.4 | **48.0** | **37.7** |
| | All tokens + PCD§ | 33.1 | 16.6 | 45.9 | 36.1 |
| | Random 10% | 32.2 | 16.7 | 46.7 | 36.4 |
| | SemVID 10% | 29.1 | 13.7 | 42.5 | 31.0 |
| | Coverage 10% | 33.2 | **17.7** | 47.5 | 37.1 |
| Joint, timeline | All tokens | 34.1 | **11.0** | 30.4 | 20.8 |
| | All tokens + PCD | 33.1 | 10.1 | 33.3 | **26.4** |
| | Random 10% | 29.8 | 9.5 | 30.5 | 20.1 |
| | SemVID 10% | 28.1 | 7.9 | 28.6 | 17.2 |
| | Coverage 10% | 33.2 | **11.0** | 31.7 | 21.7 |
| | Coverage 10% + PCD† | 32.2 | 10.7 | **33.4** | 24.7 |
| | Random 30% | 33.9 | 10.2 | 31.3 | 21.2 |
| | Coverage 30% | **34.2** | 10.9 | 30.2 | 21.3 |
| Joint, S-EMBER prompt | All tokens | 38.7 | 13.2 | 35.1 | 30.4 |
| | All tokens + PCD | **40.0** | 13.4 | 33.4 | 28.9 |
| | Random 10% | 39.5 | 12.8 | 34.4 | 29.7 |
| | Coverage 10% | 38.9 | **13.5** | **35.9** | **30.6** |
| | Coverage 10% + PCD | 39.1 | 13.2 | 33.6 | 28.2 |

ing 0. At 30%, Coverage is lossless (+0.5 Acc@GQA, −0.2 mIoU, n.s.), is +1.2 Acc@QA above random selection (n.s.), and grounds better than SemVID (+1.8* mIoU). When frames are dense, then, where tokens are kept matters most for the answer and less for the interval, as on Charades-STA (Sec. 6.7).

The 5% rows still use the earlier timestamp rule. Under that rule, Coverage leads random selection on Acc@QA by +6.8* with Qwen3-VL-8B and +5.6* with TimeLens-8B. At 10%, however, the fixed rule raised both selectors by about 2.3 mIoU* and 2.4 Acc@GQA*, so we do not interpret the 5% differences until the reruns finish. SemVID at a nominal 5% is also not budget-matched to Coverage at 5%, because it keeps about three times as many visual tokens. **[TODO: 5% Coverage, random and global top-$k$ with the fixed timestamp rule, both models; TimeLens-8B SemVID and 30%.]** We use the two-query protocol. Asking for the answer and the interval in one reply, as in prior zero-shot reports [16], lowers Acc@GQA by 16.5* for Qwen3-VL-8B and 9.2* for TimeLens-8B. The drop comes mostly from the time values, which the model then writes as minutes:seconds or copies from the prompt's example. Absolute numbers are therefore comparable only within a protocol.

**Checks and a matched-budget comparison.** Table 13 gives the floors of NExT-GQA on a subset. Without video, accuracy falls by 32.3* points and grounding drops to 7.2 mIoU; the options alone still give half of the answers. Shuffling the frames leaves accuracy unchanged and lowers mIoU by 6.7*. Shuffling only the timestamps also leaves accuracy unchanged (+0.4), but lowers Acc@GQA by 12.6* and mIoU by 10.9*. The model reads the interval from the timestamps, but it does not need them for the answer. Keeping 100% of the tokens through either selector reproduces all 538 unpruned replies exactly. Batch size 1 instead of 8 leaves 502 of 538 replies unchanged and no metric significantly different.

At the same number of visual tokens, Coverage leads SemVID on every metric. At 3.4% the margins are +13.0* Acc@QA, +9.3* Acc@GQA, +7.7* mIoP and +8.6* mIoU. At 24.9% they are +4.1*, +5.2*, +3.6* and +3.1*, where the Coverage 30% numbers come from the full run restricted to these questions. The fixed timestamp rule matters only for low-budget questions. On the 80 questions whose 10% budget is about 281 tokens, the earlier rule left Coverage at 2.4 mIoU, with 85% of its intervals starting at 0 s. The fixed rule gives 32.4.

**Open-ended answers.** Without the options (Tab. 14), answering becomes far harder (33.5 *vs.* 79.7 Acc@QA), while grounding is unchanged because the grounding query is the same. Our judge accepts the ground-truth answer for all

1,005 questions and agrees with letter correctness on 99.2% of the multiple-choice answers. Without pruning, Qwen3-VL-8B reaches 17.4 Acc@GQA zero-shot. TOGA [23], which is trained for open-ended grounded QA and judged with GPT-3.5, reaches 21.4. Coverage at 10% keeps the answers in both protocols, at $-0.3$ and $-0.9$ Acc@QA against all tokens (n.s.; cross-cluster pairs). In the two-query protocol random selection is close as well (Coverage $+1.0$ Acc@QA and Acc@GQA, n.s.), and only SemVID is clearly worse. Coverage leads it by $+4.2^*$ Acc@QA, $+4.0^*$ Acc@GQA and $+6.2^*$ mIoU. In the joint timeline protocol, random selection does lose the answers ($-4.4^*$ against all tokens). There Coverage leads it by $+3.5^*$ Acc@QA and $+1.5^*$ mIoU, and leads SemVID by $+5.2^*$ Acc@QA and $+3.2^*$ Acc@GQA.

The joint timeline answer grounds far worse than the separate query (20.8 *vs*. 37.7 mIoU). Its listed moments start where the subject is first visible, so 81% of its intervals start at 0 s, against 19% in the ground truth. In this format PCD raises mIoU, by $+5.6^*$ on all tokens and $+4.3^*$ on Coverage with the earlier rule, but not Acc@GQA ($-0.9$ and $+0.1$, n.s.). In the two-query protocol PCD slightly lowers grounding ($-1.5^*$ mIoU on all tokens, $-1.2^*$ on Coverage with the earlier rule) and leaves Acc@GQA unchanged ($-0.5$ and $-0.4$, n.s.). This matches what we saw with the one-line prompt on QVHighlights.

**The official S-EMBER prompt on NExT-GQA.** The official S-EMBER prompt asks for the answer and the interval without a list of observations. With it, the uncompressed model both answers and grounds better than with the timeline prompt ($+4.6^*$ Acc@QA, $+9.5^*$ mIoU). Under this prompt, Coverage and random selection at 10% are both level with all tokens ($+0.3$ and $-0.6$ mIoU, n.s.). PCD raises accuracy slightly on all tokens ($+1.3^*$) but lowers mIoU ($-1.5^*$ on all tokens, $-2.5^*$ on Coverage). Most of this loss is a format effect. PCD more than doubles the number of answers that write times as minutes:seconds, from 22 to 58 on all tokens and from 24 to 89 on Coverage, and the official parser scores such times as missing. In every joint run, 67–81% of the intervals still start at 0 s.

**TimeLens-8B, open-ended.** For TimeLens-8B we have grounding results only: the answers are not yet judged, and the 10% runs use the earlier timestamp rule. In the joint timeline format it grounds far better than Qwen3-VL-8B (36.3 *vs*. 20.8 mIoU, $+15.5^*$), and only 22–30% of its intervals start at 0 s, against 59–81% for Qwen3-VL-8B. Here PCD lowers grounding, by $-4.4^*$ mIoU on all tokens, $-4.5^*$ on Coverage and $-1.5^*$ on random tokens. The replies show why. TimeLens-8B often ends with an answer and no `Time:` line, and PCD adds run-on or truncated replies. Replies without a parsable interval, which are scored as IoU 0, therefore rise from 37 to 112. In the two-query protocol, Coverage and random tokens at 10% are level ($+0.1$ mIoU), and both lose about 2 mIoU$^*$ against all tokens. **[TODO: judge the TimeLens-8B open-ended answers; fixed-rule reruns of the PCD 10% rows.]**

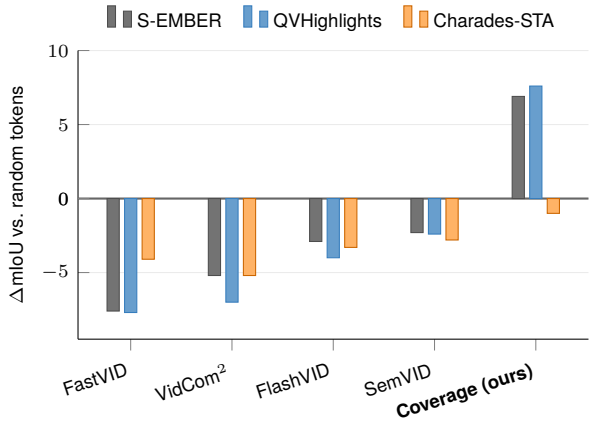



Figure 4. **QA pruners fall below random selection on grounding.** Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets (all significant except SemVID on QVHighlights). Coverage gains on the two long-video datasets and is level with random selection on 30 s Charades videos.

On NExT-GQA, then, the contribution of the method is question-driven token selection. PCD helps grounding only where the answer format lets the prior choose the times, as in the joint timeline answer of Qwen3-VL-8B, and even there it does not raise Acc@GQA. Where it disturbs the answer format, as with TimeLens-8B or with minutes:seconds under the official prompt, it lowers the scored grounding.

### 6.3. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on three of the four datasets. On S-EMBER the losses range from $-2.3$ to $-7.6$ mIoU, all significant. On QVHighlights they range from $-2.4$ to $-7.7$, with only SemVID not significant, and on Charades-STA from $-2.8$ to $-5.2$, all significant. On ActivityNet at 10%, the input-level pruners are above random, which we discuss in Sec. 6.7. These gaps are not seed noise, because random selection varies little across seeds ($26.0 \pm 0.6$ mIoU over three seeds on S-EMBER). The ordering follows coverage (Tab. 1). Methods that empty frames most aggressively lose the most, and those that spread tokens lose the least. Accuracy, by contrast, barely moves (Tab. 6). FastVID's accuracy is within noise of random selection ($-2.1$, n.s.), while its mIoU is 7.6 lower. These pruners were designed to preserve answers, and they do, but they discard the timing. PCD helps every pruner ($+7$ to $+12$ mIoU) but does not close the gap to BOCC.

Table 15. **Component ladder** (mIoU / R@0.5) with 10% of tokens; PCD: prior-contrastive decoding.

| Configuration | S-EMBER | QVHighlights | ActivityNet |
|---|---|---|---|
| Uniform, random tokens | 25.6 / 24.6 | 26.9 / 25.0 | 22.9 / 18.6 |
| BOLT, random tokens | 29.4 / 29.5 | 28.7 / 28.0 | – |
| Coverage (uniform frames) | 31.7 / 32.4 | 33.7 / 33.0 | 24.6 / 21.6 |
| BOLT + Coverage | 32.5 / 34.1 | 34.5 / 35.6 | 25.4 / 21.2 |
| BOLT + Coverage + PCD | **39.3 / 40.8** | **48.8 / 54.8** | **34.7 / 32.4** |

Table 16. **Design choices of Coverage** (S-EMBER, greedy decoding). Frames: U = uniform. Δ *vs.* the equal quota with question top-$k$ on the same frames.

| Token selection | Frames | mIoU | Δ |
|---|---|---|---|
| **Equal quota, question top-$k$** | BOLT | 32.5 | – |
| Quota ∝ SigLIP relevance (floor 50%) | BOLT | **33.0** | +0.5 |
| Quota ∝ SigLIP (floor 25%) | BOLT | 32.6 | +0.1 |
| Quota ∝ SigLIP + attention | BOLT | 32.9 | +0.4 |
| Quota ∝ attention mass | BOLT | 32.1 | −0.4 |
| One token set shared by all layers | BOLT | 31.9 | −0.6 |
| **Equal quota, question top-$k$** | U | 31.7 | – |
| Quota ∝ SigLIP relevance | U | 31.4 | −0.2 |
| Equal quota, sampling instead of top-$k$ | U | 30.0 | −1.7 |

## 6.4. Analysis of Coverage

**Contribution.** Table 15 adds the components one at a time. On the memory side, Coverage gives the largest gain over random tokens, $+6.1^*$ mIoU on S-EMBER and $+6.8^*$ on QVHighlights. BOLT adds $+0.8$ to $+0.9$ on top, which is not significant on any dataset. Once tokens are chosen by the question, the choice of encoded frames matters less. **[TODO: Coverage + PCD without BOLT on all datasets.]**

**Design choices.** Table 16 varies one choice at a time. If there is a quota, an equal one works as well as any weighted one. We tried five relevance-weighted variants, using SigLIP similarity, attention mass or both, with floors of 25–50%. All are within $\pm 0.5$ mIoU of the equal quota and none is significant (the uniform-frame SigLIP variant was our pre-registered primary test). Quotas that follow raw attention mass get pulled by attention sinks and recency bias, so that a single frame received up to 227 of about 2,140 kept tokens. The score itself should come from the actual question. HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without one. Finally, deterministic top-$k$ beats sampling from a coverage-floored density (31.7 *vs.* 30.0). A contrastive token score that subtracts the attention of a generic sentence, meant to cancel sinks, does not help.

**Is the quota needed?** For the generic HERMES score, the equal quota is worth $+5.3^*$ mIoU, because that score concentrates on the end of the window and empties earlier frames (Tab. 1). With the actual question's attention, the quota was not needed in either of the two settings we tested. There, a global top-$k$ of the same per-layer attention, with no quota, matches the equal quota. On S-EMBER under the official prompt the difference is $-0.1$ mIoU (interval $[-1.5, +1.3]$, Tab. 22). On QVHighlights with BOLT frames and the one-line prompt it is $-0.0$ ($[-0.9, +0.7]$) on the 584 queries run so far, where the global top-$k$ is $+1.8^*$ above random tokens. In these settings, the gain of Coverage over random selection comes from scoring tokens by the question. The equal quota acts as a guarantee that every anchor survives, which matters for scores that collapse onto a few frames. **[TODO: global top-$k$ without quota under the timeline prompt on S-EMBER and on the complete QVHighlights set (queued); share of frames left empty by it.]**

Table 17. **Per-layer *vs.* shared token sets** (S-EMBER, 10%). Δ = shared − per layer.

| Method | Per layer | Shared | ΔmIoU |
|---|---|---|---|
| Random tokens | 25.6 | 26.3 | +0.7 |
| Random + PCD | 35.0 | 35.8 | +0.7 |
| BOLT + Coverage | 32.5 | 31.9 | −0.6 |
| BOCC | **39.3** | **37.8** | −1.5 |

**Per-layer selection.** Table 17 compares one token set per layer with a single set shared by all layers. Sharing does not change random selection, so per-layer diversity is not why random selection beats input-level pruners. For BOCC, sharing costs $-1.5^*$ mIoU. Different layers attend to different tokens, and question-driven selection benefits from letting each layer choose its own.

## 6.5. Analysis of PCD

**PCD on other memories.** With reasoning-style answers, PCD can be added to any memory. Table 18 applies it to eight: three random budgets, the uncompressed model, HERMES and stratified pruning, and two HERMES streaming memories. PCD improves mIoU significantly on all eight ($+4.6$ to $+8.0$) and GQ@0.5 on six. It roughly halves the early starts everywhere, and its gains are largest where the memory is smallest. With 10% of the tokens and random selection, it already beats the uncompressed model (32.6 *vs.* 28.0). PCD also transfers to FastVID, the pruner that loses the most coverage, where it adds $+7.6^*$ mIoU and cuts the leak from 38% → 12%.

**Choice of counterfactual.** Table 19 changes only what the memory is contrasted against. The no-video input works best because it isolates the prior, and contrasting against it removes the prior (early starts 38% → 19%). Noised frames, as in VCD, still show part of the video *and* still pro-

Table 18. **PCD on eight memories** (S-EMBER, Acc. / mIoU / GQ@0.5; earlier implementation of PCD). Plain PCD: on all tokens with constant $\alpha$=0.5; PCD: confidence-adaptive strength on time values only (ours). $\Delta$ PCD *vs.* greedy. Best and second best per row.

| Memory | Greedy | Plain PCD | PCD | ΔmIoU | ΔGQ@0.5 | Early starts |
|---|---|---|---|---|---|---|
| Random 5% | 15.8 / 23.6 / 5.9 | 15.4 / 29.2 / 9.1 | **16.4 / 31.5 / 9.9** | +8.0* | +4.0* | 39% → 20% |
| Random 10% | 15.2 / 25.8 / 7.2 | **16.2** / 32.1 / **10.1** | **16.2 / 32.6 / 10.1** | +6.8* | +2.9 | 38% → 19% |
| Random 25% | 17.7 / 28.2 / 8.2 | **18.9** / 33.0 / 11.8 | **18.9 / 34.9 / 13.3** | +6.7* | +5.1* | 37% → 21% |
| All tokens | 18.5 / 28.0 / 8.6 | 19.8 / 31.5 / 11.2 | **21.5 / 32.7 / 13.3** | +4.6* | +4.6* | 41% → 23% |
| HERMES 10% | 13.1 / 18.3 / 4.0 | **15.4** / 22.6 / **7.2** | 13.3 / **24.1** / 5.9 | +5.7* | +1.9 | 56% → 31% |
| Stratified 10% | 14.1 / 23.7 / 6.1 | 14.5 / 29.9 / 8.8 | **15.8 / 31.0 / 10.5** | +7.3* | +4.4* | 36% → 17% |
| HERMES streaming 4k | 13.5 / 21.6 / 4.0 | **17.1** / 27.4 / **8.2** | 15.4 / **28.6** / **8.2** | +6.9* | +4.2* | 45% → 24% |
| HERMES streaming 6k | 14.3 / 22.6 / 4.8 | 14.7 / 28.1 / 7.6 | **15.4 / 29.8 / 8.4** | +7.2* | +3.6* | 46% → 24% |
| Mean of 8 | 15.3 / 24.0 / 6.1 | 16.5 / 29.2 / 9.3 | **16.6 / 30.7 / 10.0** | +6.7 | +3.9 | |

Table 19. **What to contrast against** (S-EMBER, random 10%).

| Counterfactual | mIoU | GQ@0.5 | Early |
|---|---|---|---|
| None (greedy) | 25.8 | 7.2 | 38% |
| **No video (ours)** | **32.1** (+6.2*) | **10.1** (+2.9*) | **19%** |
| Noised frames (VCD [30]) | 26.3 (+0.5) | 7.8 (+0.6) | 30% |
| Evidence-against rule, $\alpha$=1 / 2 | 26.5 / 27.3 | – | – |
| Attention gain (PAI-style) | +1.3 (n.s.) | +0.8 | 42% |
| Instruction "do not assume early" | 25.7 | 6.9 | 37% |

Table 20. **Time scope and adaptive strength** (S-EMBER, Acc. / mIoU / GQ@0.5). Top: PCD on all tokens *vs.* time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window. Top: best per row.

| Memory | All tokens | Time values only |
|---|---|---|
| Random 10% | **16.2** / 32.1 / 10.1 | 16.0 / 32.1 / **10.3** |
| Random 25% | **18.9** / 33.0 / 11.8 | 17.7 / **34.0** / 11.8 |
| All tokens | 19.8 / 31.5 / 11.2 | **20.8 / 32.9 / 12.4** |
| HERMES 10% | **15.4** / 22.6 / 7.2 | 14.7 / **24.3** / **7.4** |
| Streaming 4k | **17.1** / 27.4 / 8.2 | 16.2 / **29.2** / **9.3** |
| *Early-evidence questions (78), all tokens memory* | | |
| Greedy | 21.8 / 49.2 / 16.7 | |
| PCD, constant $\alpha$ | 16.7 / 55.3 / 12.8 | |
| PCD, adaptive $\alpha$ | **23.1 / 56.9 / 19.2** | |

duce the prior. The two runs then disagree mainly about content, and little of the prior is removed (+0.5 mIoU). Permuted timestamps also leave the prior in place. A one-sided rule that only penalises tokens the video argues against is worse than PCD ($-5.6^*$ and $-4.8^*$). Without video the model is about 99% sure of "0", so even a well-supported "0" looks like evidence against. Boosting visual attention during decoding, or stating the bias in the instruction, does not remove the prior either.

**Time scope and adaptive strength.** Contrasting every token also changes words that have nothing to do with time.

Table 21. **Decision rules replayed at the first time value** (S-EMBER, random 10%). Share of answers starting at "0".

| Rule | Early evidence ↑ | Other (leak) ↓ |
|---|---|---|
| Greedy | **69%** | 16% |
| PCD, $\alpha$=0.5 | 46% | **6%** |
| PCD, adaptive | 51% | 7% |
| Evidence-against, $\alpha$=4 | 51% | 9% |
| Skewed or tempered prior | 50–55% | 8–11% |
| Best threshold (uses gold) | 46% | **6%** |

The answer text changed in 85% of questions, and lists of more than eight moments rose from 12 to 56. Over-counting rose from 14% to 34% of counting questions. Restricting PCD to time values keeps the grounding gain and disturbs the text less (Tab. 20, top). PCD also risks hurting questions whose evidence really starts at 0 s. With a constant strength, their accuracy on the uncompressed memory falls from 21.8 to 16.7. The adaptive strength fades when the model is confident, and with it accuracy rises to 23.1 instead (Tab. 20, bottom; $+6.4^*$ *vs.* constant). The adaptive rule also gives the best *discrimination*, defined as early starts on early-evidence questions minus early starts on the others. On the all-tokens memory it reaches 56.4, against 49.6 for greedy decoding.

**Other decision rules.** We logged both distributions at the first time value of every answer and replayed other decision rules offline (Tab. 21). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones, so PCD is close to what decoding alone can do. In the early-start errors that remain, the memory itself is unsure, and fixing them would take more evidence in the memory.

**Is PCD more than a longer interval?** PCD lengthens predicted intervals (median predicted/gold length 0.57 → 0.67 on S-EMBER), and longer intervals alone raise mIoU on benchmarks with long gold intervals. The length-rescaling floor tests this directly (Tab. 5, last row). BOCC

Table 22. **Official S-EMBER prompt** (direct answer with `Answer:` and `Time:` lines; Qwen3-VL-8B, 64 frames, 10%, second cluster; grounding only, answers not yet judged). mIoU / R@0.5 with the minutes:seconds-aware parser; mIoU with the official parser and the number of answers it cannot parse.

| Method | mIoU / R@0.5 | Official mIoU | Unparsed |
|---|---|---|---|
| All tokens | [TODO: queued] | | |
| Random tokens | 27.9 / 26.1 | 26.4 | 17 |
| Coverage, uniform frames | 30.1 / 29.3 | 28.8 | **13** |
| Global question top-$k$, no quota | 30.0 / 29.3 | 28.5 | 17 |
| BOCC | **34.6 / 34.3** | **29.5** | 73 |

clearly beats rescaling on QVHighlights ($+8.6^*$) and on S-EMBER closed intervals ($+7.5^*$, Tab. 29). On these sets PCD moves the interval toward the evidence instead of only stretching it. On pooled S-EMBER, ActivityNet and Charades-STA the two are statistically tied, so there the benefit of PCD cannot be separated from a calibrated lengthening. **[TODO: start/end error decomposition on closed questions; mismatched-video and shuffled-frame negatives.]**

**Prompt format.** We do not yet have a clean PCD-only comparison on S-EMBER under the official direct-answer prompt. An earlier run that was not matched on the other settings gave $+0.4$ mIoU. Under this prompt (Tab. 22), BOCC is $+4.5^*$ mIoU above Coverage alone and $+6.8^*$ above random tokens. This gain mixes PCD with BOLT, since on the first 116 questions BOLT + Coverage without PCD already gains $+5.3^*$ over Coverage. BOCC is also $4.7^*$ below BOCC with the timeline prompt. Coverage is $+2.2$ (n.s.) above random tokens. PCD makes the model write more times as minutes:seconds, and 73 BOCC answers cannot be parsed by the official parser, against 13–17 for the greedy rows. For the same reason, PCD alone lowers mIoU on NExT-GQA under the same prompt ($-1.5^*$ and $-2.5^*$, Sec. 6.2). Until the uncompressed model has been run under this prompt, we cannot say whether BOCC improves on it. Compared with the uncompressed model in the timeline format, BOCC with the official prompt is $+6.6^*$ and Coverage $+2.1$ (n.s.). **[TODO: uncompressed model under the official prompt (queued); judged answers; PCD with and without BOLT under this prompt.]** We also tried an *event* timeline that lists start and end per occurrence (`Event: 12.0 -- 15.5 seconds`). Because it states extents, it improves the greedy baseline by $+7.4^*$ mIoU. PCD still adds $+2.1$ to $+3.3$ on top, reaching 35.0 mIoU and 12.6 GQ@0.5 at 768 tokens. The two fixes overlap. The format corrects extents, and PCD corrects the leak. BOCC uses the simpler timeline in all main results. **[TODO: event timeline with BOCC.]**

**When does PCD help?** The results so far suggest a simple rule. PCD helps in proportion to how much the video-free prior decides the time that is written. Its gain is largest where the memory says least, $+8.0^*$ mIoU for random 5% against $+4.6^*$ for the uncompressed model (Tab. 18). The adaptive strength fades when the memory makes the model confident, and so protects true early starts that a constant strength removes (Tab. 20). With a one-line answer, where the model states a single interval instead of listing observations, PCD has no effect (Sec. 6.2; those runs had no answer reserve). One qualification is needed, because PCD can also change the *form* of the answer. On NExT-GQA it lowers TimeLens-8B's grounding in the timeline format ($-4.4^*$ mIoU) by producing more replies without a parsable interval. Under the official prompt it makes Qwen3-VL-8B write more times as minutes:seconds (Sec. 6.2). In short, PCD helps when the prior chooses the times and the answer keeps its format. We state this as a hypothesis, which the per-token log-probabilities we already store can test directly. **[TODO: PCD gain against the divergence between $p_{\mathcal{M}}$ and $p_{\varnothing}$ at each written time value, across all conditions; one prediction stated before the run.]**

## 6.6. Budget

Table 23 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection stays flat or even improves as the budget *shrinks*. On S-EMBER, BOLT + Coverage rises from 31.0 at 20% to 34.6 at 5%, and BOCC from 36.8 to 39.3. Random selection and SemVID degrade over the same range. Pruning by the question thus removes distractors as well as cost. At 5%, BOCC leads SemVID + PCD by $+7.4^*$ mIoU, and at 20% by $+1.5$ (n.s.).

Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair. The top-attended tokens then carry little content, and BOCC collapses to 9.7 and 11.6. This is below random selection and below SemVID, whose cross-frame merging degrades more gracefully. What matters is the absolute number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies. One caveat applies. These runs count the kept timestamps inside the budget under the earlier timestamp rule, which on NExT-GQA left some low-budget questions (about 280 tokens) with no visual tokens (Sec. 6.2). Part of the collapse below about 300 tokens may therefore come from this accounting rather than from the equal quota. **[TODO: QVHighlights 5% and 10% at 64 frames with the fixed timestamp rule (queued).]**

## 6.7. Fewer frames or fewer tokens per frame?

The collapse at small budgets comes from how the budget is spent, and the equal quota is not at fault in itself. The timestamp-accounting caveat of Sec. 6.6 still applies here, since fewer frames also means fewer timestamps inside the

Table 23. **Budget sweep at 64 frames** (mIoU). The second row gives the number of kept tokens per memory.

| | S-EMBER | | | QVHighlights | | | ActivityNet | | |
|---|---|---|---|---|---|---|---|---|---|
| Budget | 5% | 10% | 20% | 5% | 10% | 20% | 10% | 30% | 50% |
| Kept tokens | ∼1,070 | ∼2,140 | ∼4,280 | ∼260 | ∼520 | ∼1,040 | 120–300 | 400–900 | 600–1,500 |
| Random | 23.9 | 25.6 | 27.4 | | 26.9 | | 17.8 | 22.9 | 23.5 |
| SemVID | 20.9 | 23.3 | 26.5 | | 24.5 | | 21.2 | 22.8 | 22.3 |
| R + C (no PCD) | 34.6 | 32.5 | 31.0 | | 34.5 | | 11.3 | 25.4 | 25.0 |
| SemVID + PCD | 31.9 | 34.8 | 35.3 | | 34.8 | | **26.8** | 30.2 | 31.4 |
| Random + PCD | 34.5 | 35.0 | 36.5 | | 37.8 | | 21.5 | 31.4 | 32.2 |
| BOCC | **39.3** | **39.3** | **36.8** | 9.7 | **48.8** | 49.4 | 11.6 | **34.7** | **34.4** |

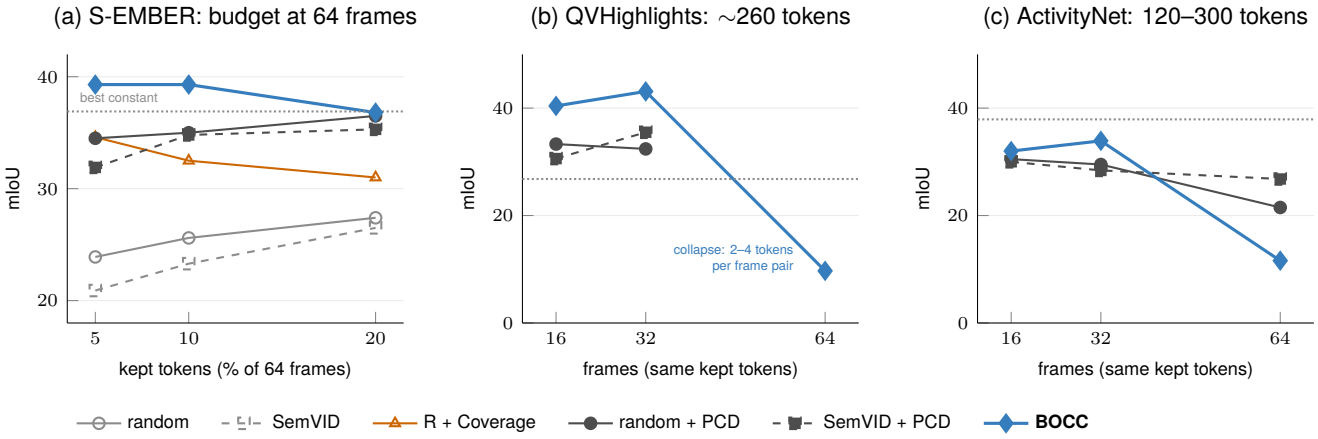



Figure 5. **Budget and how to spend it. (a)** At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; BOCC leads at every budget that leaves at least about eight tokens per frame pair, and collapses below it (QVHighlights 5%, ActivityNet 10%; these runs use the earlier timestamp rule, Sec. 6.6). **(b, c)** At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and BOCC collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9). Dotted grey: best video-free constant.

budget. In Tab. 24, every row of a block keeps the same number of tokens, and only the split between frames and tokens per frame changes. With 32 frames instead of 64, each frame pair keeps twice as many tokens, and BOCC recovers. On QVHighlights it goes from 9.7 to 43.1 mIoU. On ActivityNet it goes from 11.6 to 33.9, which beats SemVID + PCD by +5.5 and nearly matches its own 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little, because past a point temporal coverage becomes the bottleneck. The baselines also gain from fewer frames, but less. On Charades-STA, whose videos last 30 s, fewer frames do not help any method. We therefore recommend the rule of Sec. 4.1. Keep at least about eight tokens per frame pair, and reduce $F$ when the budget does not allow it. At equal tokens, more frames with fewer tokens each do not help either. With 128 frames at 5% on S-EMBER, the change is +1.2 mIoU (n.s.) without PCD and −3.6* with it, with 74–78 unparseable answers.

## 6.8. Comparison with gold-informed upper bounds

Table 25 gives the memory information it should not have. On Qwen3-VL-8B, BOCC beats perfect token selection at the same budget without using any gold information. Over the oracle selector with PCD, it gains +4.4* mIoU and +4.0* GQ@0.5. Knowing which tokens to keep does not stop the prior, and keeping only the gold frames' tokens starves the rest of the timeline. The strongest oracle sees only the evidence and also uses PCD. Against it, BOCC is level on grounding (−2.2 mIoU and −1.5 GQ@0.5, both n.s.) but answers worse (−5.3* accuracy). Even the gold window leaks 45%, so with every frame inside the evidence the model still often starts near 0 s. On the gold window, PCD lowers the leak to 35% and adds +5.5* mIoU, but it does not change the answers (−2.3, n.s.). For Qwen3-VL-8B, then, more evidence raises accuracy, while PCD fixes the time values.

For TimeLens-8B (Tab. 26), the bounds are much higher and BOCC is further from them. Over the oracle selector

Table 24. **Spending a fixed token budget on fewer frames** (mIoU / R@0.5, all with PCD). Within each block every row keeps the same number of tokens: $F$ frames at keep ratio $\rho \cdot 64/F$. Best and second best per row.

| Dataset (kept tokens) | Frames × keep ratio | Random + PCD | SemVID + PCD | BOCC |
|---|---|---|---|---|
| QVHighlights (∼260) | 64 × 5% | – | – | 9.7 / 6.6 |
| | 32 × 10% | 32.4 / 29.2 | 35.5 / 33.6 | **43.1 / 46.6** |
| | 16 × 20% | 33.3 / 32.4 | 30.6 / 28.0 | **40.4 / 41.0** |
| ActivityNet (120–300) | 64 × 10% | 21.5 / 18.2 | **26.8 / 23.6** | 11.6 / 7.6 |
| | 32 × 20% | 29.5 / 26.8 | 28.4 / 25.8 | **33.9 / 32.0** |
| | 16 × 40% | 30.5 / 26.8 | 30.0 / 27.6 | **32.0 / 29.8** |
| Charades-STA (∼10% of 64) | 64 × 10% | **26.2 / 24.0** | 21.7 / 19.2 | 23.3 / 21.2 |
| | 32 × 20% | **24.6 / 22.0** | 20.8 / 16.0 | 23.6 / 21.0[§] |

[§] Coverage on uniform frames; BOLT is skipped on Charades-STA.

Table 25. **Upper bounds that use the gold interval** (S-EMBER, Qwen3-VL-8B, 475 questions; Acc. and GQ@0.5 from judging session B; minutes:seconds-aware parser). Oracle selector: the gold frames' tokens are kept first. Gold window: all 64 frames inside the gold interval (batch size 3). Leak: intervals starting in the first 5% of the window, on the 397 questions whose evidence starts later.

| Memory | Gold | Acc. | mIoU | R@0.5 | GQ@0.5 | Leak |
|---|---|---|---|---|---|---|
| Random 10% | – | 17.1 | 25.6 | 24.6 | 7.2 | 29% |
| Random 10% + PCD | – | 20.0 | 35.0 | 33.9 | 12.0 | 14% |
| Oracle selector 10% | ✓ | 18.9 | 27.9 | 25.7 | 9.3 | 33% |
| Oracle selector 10% + PCD | ✓ | 18.5 | 34.9 | 35.2 | 9.5 | 16% |
| BOCC 10% | – | 21.9 | 39.3 | 40.8 | 13.5 | **11%** |
| All tokens | – | 19.6 | 28.0 | 26.7 | 8.8 | 33% |
| Gold window | ✓ | **29.5** | 36.1 | 38.1 | **15.8** | 45% |
| Gold window + PCD | ✓ | 27.2 | **41.5** | **43.2** | 14.9 | 35% |

Table 26. **Gold-informed upper bounds for TimeLens-8B** (S-EMBER, 475 questions; all Acc. and GQ@0.5 from judging session C, Sec. 5). Settings as in Tab. 25; all tokens at batch size 1. No oracle selector without PCD was run.

| Memory | Gold | Acc. | mIoU | R@0.5 | GQ@0.5 |
|---|---|---|---|---|---|
| Random 10% | – | 18.5 | 28.2 | 26.7 | 9.1 |
| Random 10% + PCD | – | 19.8 | 38.4 | 39.4 | 11.8 |
| Oracle selector 10% + PCD | ✓ | 21.7 | 43.3 | 45.3 | 14.3 |
| BOCC 10% | – | 26.5 | 43.9 | 46.9 | 17.1 |
| All tokens | – | 19.8 | 33.1 | 34.1 | 11.6 |
| Gold window | ✓ | 28.2 | 46.6 | 46.7 | 19.2 |
| Gold window + PCD | ✓ | **36.2** | **54.9** | **57.1** | **25.9** |

with PCD, BOCC is level on grounding, with +0.5 mIoU and +2.7 GQ@0.5 (n.s.). Its accuracy is +4.8 higher, with an interval lower bound of 0.0. The gold window alone is +13.2* mIoU above all tokens (both at batch size 1). Adding PCD on the gold window raises mIoU by +8.3*, and it also raises accuracy (+8.0*) and GQ@0.5 (+6.7*). BOCC stays below this bound by 11.0* mIoU, 9.7* accuracy and 8.8* GQ@0.5. So for the grounding-tuned model, PCD and evidence work together. With the right frames, contrasting the time values also improves the answers. The gap to the gold window shows how much a better choice of frames could still add for this model.

### 6.9. Analysis of BOLT

The SigLIP 2 index does find the evidence. Over the seconds before each question, the AUROC of evidence seconds against the rest is 0.663. The top-ranked second lies in the gold interval for 63% of questions (chance 36%), and for 41% of location questions (chance 13.5%). As a result, BOLT frames put 40% of the memory inside the evidence, against 36% for uniform frames. This helps random tokens (+3.8* mIoU). Once tokens are chosen by the question, however, BOLT adds only +0.8 to +0.9 (n.s.), because both stages focus the memory on the question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666). This is expected for a retriever trained with a contrastive loss, whose score already estimates pointwise mutual information [45]. Re-weighting the token budget of uniform frames by the same relevance adds +1.3 (n.s.). The choice of encoded frames matters more than how the budget is split among fixed frames.

### 6.10. What did not work

Table 27 collects the ideas that gave little or nothing. Taken together, they suggest that at a fixed budget, redistributing evidence by zooming, re-weighting frames or unequal quotas is close to its ceiling. Large gains came only from removing the prior where it acts (PCD), or from selecting tokens by the actual question in every layer, with or without the equal quota (Sec. 6.4).

### 6.11. Where answers still fail

Table 28 shows that PCD works where it was aimed. Intervals entirely before the gold one are the signature of the

Table 27. **Negative results** (S-EMBER; ΔmIoU against the stated reference).

| Idea | Reference | ΔmIoU |
|---|---|---|
| VCD-style noised-frame contrast | greedy | +0.5 |
| Permuted-timestamp contrast | greedy | +0.9 |
| Evidence-against decoding rule | PCD | −5.6* |
| Attention gain while decoding | greedy | +1.3 |
| Bias stated in the instruction | greedy | −0.1 |
| Relevance-guided frame zoom | greedy | −1.5 |
| Zoom inside the gold window | greedy | +2.3 |
| Unequal per-frame quotas (5 variants) | equal quota | −0.4 to +0.5 |
| Contrastive token score | question top-$k$ | −1.2 to +0.5 |
| Prior-subtracted retrieval | SigLIP | AUROC ≤ +0.003 |
| 128 frames at the same budget | 64 frames | +1.2 / −3.6* |

Table 28. **Error types** on S-EMBER, pooled over eight memories, greedy *vs*. PCD (earlier implementation of PCD).

| Outcome | Greedy | PCD | Δ |
|---|---|---|---|
| IoU ≥ 0.5, answer correct | 6.1% | 9.9% | +3.8* |
| IoU ≥ 0.5, answer wrong | 16.3% | 20.7% | +4.4* |
| Partial overlap (IoU < 0.5) | 17.0% | 15.7% | −1.3 |
| Inside gold, too short | 22.2% | 18.5% | −3.7* |
| Entirely *before* gold | 33.1% | 18.9% | −14.2* |
| Entirely after gold | 2.7% | 5.7% | +3.0* |
| Covers gold, too long | 2.2% | 2.5% | +0.3 |
| No answer / unparsed | 0.5% | 8.1% | +7.6* |

prior. Under greedy decoding they are the largest error type, and with PCD they fall to less than a fifth of answers. The remaining failures point to four limits. (i) Wrong answers on good intervals. Twice as many answers are well grounded but wrong as are grounded and right. Counting answers do not use the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) Recent events. Of the intervals that still end before the gold one, 81% have evidence in the second half of the window, and the model picks an earlier similar event instead. When the evidence lies in the last third of the window, GQ@0.5 is 4.4. (iii) Sparse evidence. With fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for any method, against 15% with 16 or more. No decoding rule can recover evidence that is not in the input. (iv) Label ambiguity. On the 98 questions whose annotators disagree (mean pairwise IoU < 0.6), every method scores near zero.

## 7. Limitations

**Answer format.** The main tables use a reasoning-style timeline prompt. On single-moment grounding this prompt costs the uncompressed model about 30 mIoU compared with the standard one-line prompt, and with it Qwen3-VL-8B stays below the best constant interval on ActivityNet Captions and Charades-STA. Our gains over the uncompressed model on QVHighlights, ActivityNet and Charades-STA (Tab. 4) therefore hold relative to this format. The comparison under the one-line prompt is in Tab. 9, and for the full Charades-STA and QVHighlights sets in Tab. 10. We have not yet run the full ActivityNet set. On S-EMBER, BOCC beats the uncompressed model when both answer in the timeline format. Under the official direct-answer prompt, BOCC reaches 34.6 mIoU (grounding only, not yet judged). That is 4.7* below BOCC with the timeline prompt and +4.5* above Coverage alone (Tab. 22). We have not yet reported the uncompressed model under this prompt, so we cannot say whether BOCC improves on it. **[TODO: S-EMBER without pruning under the direct-answer and one-line prompts; judged answers under the direct-answer prompt.]**

**Where the prior is measured.** All our measurements of the video-free prior (Sec. 3.3) use the timeline format. With a one-line answer, PCD has no effect (in runs without the answer reserve), which suggests that the prior decides few or none of the times written in that format. We have not yet measured the no-video answers in it. **[TODO: no-video answers under the one-line and direct-answer prompts.]**

**What drives Coverage.** The equal quota guarantees that every timestamp survives, and it helps scores that concentrate on a few frames, such as the HERMES probes (+5.3*). With attention from the actual question, however, a global top-$k$ without the quota matched it in both settings we tested, S-EMBER with the official prompt and a partial run on QVHighlights with BOLT frames. Our evidence that grounding needs anchors in every frame therefore comes from selectors that empty frames, namely the QA pruners and the generic probe scores, and does not rest on the quota itself. **[TODO: global top-$k$ without quota under the timeline prompt; frame coverage of question top-$k$.]**

**Minimum tokens per frame.** Equal quotas fail when a frame pair keeps only 2–4 tokens. Spending the budget on fewer frames avoids this (Sec. 6.7), but we derived our rule of at least about eight tokens per frame pair from those same results, and it still needs validation on held-out data. The small-budget runs also use the earlier timestamp rule, which left some NExT-GQA questions with no visual tokens, so part of the collapse may be an accounting effect. **[TODO: small-budget runs with the fixed timestamp rule.]**

**Short, densely sampled videos.** On Charades-STA, uniform frames are less than a second apart, and on NExT-GQA 64 frames already cover a 40 s video. On such videos question-driven selection improves grounding only slightly over random selection: by +0.5* and +0.7* mIoU on Charades-STA, and by +0.6 and +1.1 mIoU (n.s.) on NExT-GQA at 10%. Its benefit there shows up in the answers instead. On NExT-GQA at 10% it keeps the accuracy of the uncompressed model, while random selection loses

about four points. The 5% NExT-GQA results use the earlier timestamp rule and are being rerun. On these videos PCD improves the interval only in the timeline format and only for Qwen3-VL-8B, and even there it does not raise grounded accuracy (Tab. 14). **[TODO: 5% NExT-GQA with the fixed timestamp rule.]**

**Where PCD hurts.** PCD is meant to change which time values are written, but it can also change how they are written. On NExT-GQA it lowers TimeLens-8B's timeline grounding by $-4.4^*$ mIoU, because more replies end up without a parsable interval. Under the official prompt it makes Qwen3-VL-8B write more times as minutes:seconds, which the official parser scores as missing. In deployment one would need a parser that accepts both forms, or a format check on the contrasted reply.

**Upper bounds and the front end.** BOCC remains well below the gold-window upper bound of TimeLens-8B with PCD, by $-11.0^*$ mIoU and $-9.7^*$ accuracy. For Qwen3-VL-8B it is below the gold window in accuracy ($-5.3^*$). With BOLT frames, most of the gain over a low-resolution input comes from frame selection and resolution, and random tokens on the same frames get it too. We also do not count the cost of the SigLIP index.

**Not a streaming memory.** We rebuild the memory for each question from the frames that precede it. The memory is causal, but the full memory is prefilled once per question, so peak memory and time-to-first-token match those of the uncompressed model. BOCC reduces the memory the decoder reads and leaves the cost of building it unchanged. PCD also decodes a second, prompt-only row at every step. **[TODO: cost table; amortisation over several questions per video; write-once memory re-ranked at question time.]**

**Scope of the evaluation.** Both of our models come from one family. TimeLens-8B is fine-tuned from Qwen3-VL-8B, and both read time from text timestamps. Most diagnostic results use subsets of 475–500 questions. Answers are judged by a local model with the official prompt, in three judging sessions that differ by up to about one accuracy point. Some paired differences combine runs from two GPU clusters, whose repeated runs agree within one mIoU point. We chose every hyperparameter on the S-EMBER questions. The other datasets were run once with the frozen configuration, but we have not yet evaluated a held-out S-EMBER split. **[TODO: a second model family; held-out S-EMBER split.]**

## 8. Conclusion

Pruning for QA keeps what is salient, but grounding needs what is everywhere. A video LLM reads time from timestamp anchors that survive only together with their frames. Pruners built for QA empty the frames that look unremarkable, and as a result they fall below random selection on grounding. If each layer instead keeps the tokens that the actual question attends to, the timeline stays usable. With a quarter of the visual tokens, this matches the uncompressed model under the standard grounding prompt on the full Charades-STA and QVHighlights sets, and with a tenth of the tokens it keeps NExT-GQA accuracy, for both a general and a grounding-tuned model. An equal per-frame quota guarantees every anchor, although with the question's attention it was not needed in the settings we tested.

A second failure appears when the answer lists what was seen before it states the interval, as grounded QA does. Where the memory is silent, a video-free prior writes the times. Contrasting each time value with the model's own no-video prediction removes this prior, and with a tenth of the tokens BOCC then grounds better than the uncompressed model answering in the same format. Where the contrast disturbs the answer format, though, it can lower the scored grounding. The answer format itself changes single-moment grounding by about 30 mIoU. It should be reported, together with video-free floors, in every evaluation of compressed video memories.

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## A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, *e.g.* `Time: [1:45, 2:07]`. The official S-EMBER parser scores such times as missing, so we convert `h:mm:ss` and `m:ss` to seconds, for every method alike. With the official parser the method rankings do not change. The main numbers become 44.6 / 50.2 for BOCC on QVHighlights (*vs.* 48.8 / 54.8) and 33.3 / 31.0 on ActivityNet at 30% (*vs.* 34.7 / 32.4). **[TODO: full table with the official parser.]**

## B. S-EMBER closed intervals

Two thirds of S-EMBER gold intervals end at or close to the question time, so a constant interval ending at $q$ is a strong floor on the pooled set. Table 29 therefore restricts the main comparison to the *closed* questions, whose gold interval ends before $0.8q$ (33% of the questions). This shortcut does not exist there, and we re-tune the floors on this subset.

Table 29. **S-EMBER closed intervals** (10%, mIoU / R@0.5; settings as in Tab. 4). Right: paired ΔmIoU of BOCC (*: 95% paired bootstrap interval excludes 0).

| Method | mIoU / R@0.5 | BOCC *vs.* |
|---|---|---|
| *Best constant interval* | *32.1 / 29.9* | +7.6* |
| *Length rescaling of R + C* | *32.3 / 33.1* | +7.5* |
| All tokens (no pruning) | 30.0 / 27.4 | |
| Random tokens | 28.3 / 27.4 | |
| Random + PCD | 36.8 / 38.2 | |
| SemVID + PCD | 34.3 / 34.4 | |
| Coverage only | 33.0 / 35.7 | |
| **BOCC** (Coverage + PCD) | **39.7 / 43.3** | – |

## C. Checks against published numbers

Before comparing pruning methods under the TimeLens-Bench protocol, we reproduce the published unpruned scores with the TimeLens evaluation script. We also check that our own inference pipeline gives the same scores as the script on the same queries (Tab. 30).

## D. End-snapping on S-EMBER

Setting every predicted end to the question time $q$ raises every S-EMBER method to 42–46 mIoU, above the 39.3 of BOCC. This works because 67% of the gold intervals end within the last fifth of the window. On closed intervals, end-snapping hurts every method and brings it down to 22–24 mIoU, while BOCC keeps 39.7. On QVHighlights it lowers BOCC from 48.8 to 40.5. A natural extension is a decoder that decides whether an interval is still open at question time. **[TODO: open-interval decoding.]**

Table 30. **Reproduction of TimeLens-Bench** (mIoU, no pruning). Top: TimeLens script on the full sets *vs.* published. Bottom: our pipeline *vs.* the TimeLens script on the same queries (paired Δ).

| Model | Dataset (queries) | Ours | Reference | Δ |
|---|---|---|---|---|
| Qwen3-VL-8B | Charades-STA (3,363) | 48.5 | 48.3 (published) | |
| Qwen3-VL-8B | QVHighlights (1,541) | 59.6 | 59.4 (published) | |
| Qwen3-VL-8B | ActivityNet (4,500) | 47.1 | 46.8 (published) | |
| TimeLens-8B | Charades-STA (3,363) | 55.4 | 55.2 (published) | |
| TimeLens-8B | QVHighlights (1,541) | 65.7 | 65.5 (published) | |
| TimeLens-8B | ActivityNet (4,500) | 53.2 | 53.2 (published) | |
| Qwen3-VL-8B | Charades-STA (400) | 48.0 | 47.5 (script) | +0.5 |
| Qwen3-VL-8B | ActivityNet (400) | 47.1 | 45.7 (script) | +1.4 |
| Qwen3-VL-8B | QVHighlights (513) | 60.5 | 60.5 (script) | 0.0 |
| TimeLens-8B | QVHighlights (513) | 67.1 | 66.1 (script) | +1.0 |

## E. Reproduction

BOCC corresponds to the following overrides of our code, which we release with the paper:

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time
(BOLT) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```

Floors and paired intervals are computed by `docs/paper/analysis/floors.py`.